

Cover Letter for “Self-Targeting in U.S. Transfer Programs”

We thank the Editor for offering us the opportunity to revise and resubmit our paper to the *Journal of Political Economy*. We are also extremely grateful to the Editor and Reviewers for their comments.

In what follows, we first provide an outline to our main revisions, often indicating the Reviewers’ comments where we discuss the analysis in more detail. After, we proceed through the comments in the order they appear in the original reports. Our replies quote the Editor’s and Reviewers’ comments, which we place in bold. We successfully add additional tests or analysis for each requested comment. If we misunderstood any comments, we stand ready to implement the appropriate changes.

Outline of Revisions

1. Consumption. Why do we focus on consumption? Why does consumption differ from income in this population? Under what conditions is consumption a good measure of “need,” and is our analysis likely to be valid even when the relationship between consumption and need is complicated? The Editor and Reviewers 2 and 3 ask us several foundational questions of this kind.

- a. Our paper asks whether transfer recipients self-target according to marginal utility, taking lower consumption as a proxy for higher marginal utility. R2, Comment 8 questions whether marginal utility is one-to-one in consumption. First, new exhibits (p.18) show robustness of self-targeting to more complicated utility functions which could introduce non-monotonicity: age/cohort shifts, state dependence in health, consumption-leisure nonseparability, and dependence on household composition. Second, we show self-targeting on quasi-exogenous measures of need (p. 17), which suggests that our main conclusions hold even if one worries about consumption as a proxy for marginal utility (R2, Comment 7; R3, Comment 2). Third, we present new sensitivity analyses accounting for the direct effects of transfer receipt on consumption (R2, Comment 7 and R3 Comment 5).
- b. A new subsection connects self-targeting to theories of consumption (p.19). We show differences in take-up are well explained by differences in future consumption, consistent with simple life-cycle permanent income hypothesis models, rather than by borrowing constraints or precautionary saving (R2, Comments 4 and 6).
- c. We revised the literature review (p.5) to more appropriately cite research on consumption insurance from transfer programs, following the Editor and R2, Comment 4. We also justify our focus on consumption rather than income in Sections 1 and 2 (R2’s Comment 2).

2. Consumer Expenditure Survey (CEX). In their Comment 1, R2 asked why we used the PSID rather than the CEX in the paper. We have now replicated our analyses in full in the CEX, presenting these results side-by-side where possible (R2, Comment 2).

3. Writing. We rewrote Section 3.1 (pp. 11-13) with an improved discussion of our self-targeting regression and the key assumptions. The aim is to tighten the link between the empirical analysis in

Section 3 and the welfare analysis in Section 4 and be transparent about the assumptions underlying our approach. These edits aim to follow R3's Comments 1-4, 6, 8, and 12, along with R2's Comments 5 and 6. We also adopt a more cautious tone in the conclusion, per R3's suggestion (Comments 16, 20), and cut the original manuscript's Section 4 (a distributional analysis), per R1's Comment 3.

4. Measurement Error. The referees expressed additional concerns about measurement error, which motivated a number of new checks. We present some highlights here. First, we give an econometric decomposition of the possible bias from eligibility mismeasurement (Appendix C.3, responding to R3's Comment 13), and we present evidence on the likely sign and magnitude of each component. Second, a new analysis demonstrates self-targeting on consumption even if we assume substantial income mismeasurement (R2, Comment 3; R3, Comment 15). Third, we present three robustness checks that address concerns about our use of annual data (versus higher-frequency data used by transfer programs), including focusing on populations or alternative outcomes where this issue is unlikely to arise (R2, Comment 9; R3, Comment 14).

5. Additional Analysis. Beyond these, we also conduct several additional analyses requested by referees. For instance, we investigate heterogeneity across programs (R1, Comment 1); estimate MVPFs (R1, Comment 3); and rework the welfare analysis to align it as closely as possible with the main empirical content (R2, R3).

Editor

We thank the Editor again for their suggestions and support of our paper.

“I see the biggest issue as the one raised by referee 2 regarding the concern over whether you are really able to measure need using consumption as you do.”

We thank the Editor for raising this point. We have made several major revisions to the paper and its analysis in light of the Reviewer’s comments. The Reviewer’s perspective is reasonable. First, survey consumption data are imperfect, even if they are the best available for this research question. Second, distinct from measurement, consumption is not one-to-one with need in some models. On both points, however, a number of new analyses support our main finding of advantageous self-targeting on need.

Toward this end, we made the following edits (corresponding to the same numbers immediately below):

- 1) Section 3.1 (pp.11-13) has been rewritten to better motivate the focus on consumption from the standpoint of marginal utility.
- 2) Panel D of Figure 1 (p.16) now uses model-consistent values of marginal utility on the left-hand side of our self-targeting regression. The aim is to provide a “stepping stone” from the descriptive consumption-rank regressions to the full welfare analysis.
- 3) Section 3.3 (pp.18) considers the relevance of specific departures from our baseline model (in which we assume a simple monotonic mapping from consumption to marginal utility). We test empirically if we would still find self-targeting on need in more complicated models (e.g., with preference heterogeneity in consumption or state-dependence on health).
- 4) Section 3.4 (p. 17) provides evidence of self-targeting where, instead of using consumption as the outcome variable, we instead use characteristics associated with need, like low levels of education and poor health. We think it is harder to argue that all these characteristics do not capture some notion of need. This test addresses measurement issues, as well as questions about the model in which consumption proxies for marginal utility.

These tests are discussed in R2 Comments 5–8. We also provide a more affirmative defense of using consumption to proxy for need in our response to R2, Comment 2. In what follows, we quote from our responses, editing our responses slightly to unify the presentation:

- 1) We now define, explicitly and upfront, the theoretical object of ultimate interest: $[E[u_{c,1}] - E[u_{c,0}]] / E[u_c]$, which we explain as follows:

“The ideal measure of self-targeting in a given transfer is the average difference in the social marginal utilities of income (“need”) between the transfer’s recipients and eligible nonrecipients who are otherwise identical to the government...where $u_{c,1}$ and $u_{c,0}$ respectively indicate the marginal utilities of recipients and eligible nonrecipients. The

denominator, average marginal utility, expresses this difference in money-metric terms. That is, the measure in Equation 3 captures a society's estimated willingness to pay (per dollar) to redistribute resources from eligible non-recipients to recipients, under the assumption that fiscal externalities from this action are offset via lump-sum taxes on all eligible individuals."

We then emphasize the inescapable role of a model in this section. Stepping back, if one wants to ask if transfers are well-targeted on "need" (that is, on marginal utility), one requires a utility model that yields some mapping $u'(c; \theta)$ from consumption c to marginal utility u' :

"Translating the concept of marginal utility differences in Equation 3 into an empirical framework like Equation 4 involves several key choices. Many of these choices have parallels in the literature on measuring poverty and material hardship (e.g., Citro and Michael, 1995). Here we discuss four worthy of focus: the choice of consumption as a proxy for need..."

First, marginal utilities of income are unobserved, requiring a model to link the data to marginal utility. Here, we take consumption as a proxy for marginal utility. Two broad motivations for this choice are the partial insurance of household consumption against income shocks and the mismeasurement of low incomes in both survey and administrative data. This choice aligns with consumption-savings models where marginal utility is monotonic in consumption. Under this monotonicity, consumption ranks align with marginal-utility ranks in a cross-section of households. Lifetime-income ranks would also align in this sense if consumption is fully insured against income shocks. We therefore describe transfers as "advantageously self-targeted" if take-up among eligible households predicts lower ranks in the consumption or lifetime-income distributions.

We also discuss several well-known objections that economists have raised to the idea that the mapping $u'(c, \theta)$ from consumption to marginal utility is simple and monotone:

Nonetheless, imputing marginal utility from consumption has inherent limitations. Perhaps the most significant is that it is infeasible to fully capture all variation in preferences when performing this imputation. For instance, factors like aging, parenthood (Blundell and Preston, 1998), opportunity costs of time (Aguiar and Hurst, 2005), and habit formation (Chetty and Szeidl, 2016) all influence the relationship between consumption and marginal utility. Heterogeneity in labor-supply elasticities or nonseparability of labor and consumption create similar difficulties. A second set of challenges arises from measurement issues in observed consumption, including systematic underreporting of certain consumption types (Meyer and Sullivan, 2023)."

- 2) Our "stepping stone" analysis is our new "Fact 4" (pp.15-17). We quote this new text in full, and highlight the new figure (Panel D of Figure 1):

“Fact 4: Marginal utilities of income appear higher among transfer recipients than among eligible nonrecipients with similar incomes.

Panel D connects our descriptive self-targeting regressions to the welfare analysis, and specifically to the welfare benefits of self-targeting. We do so by replacing the household-rank dependent variable with a calibrated value for each household's marginal utility of income. The resulting coefficients can be interpreted as the additional value of moving a marginal dollar from a transfer's eligible nonrecipients to its recipients. To calibrate marginal utilities for each household, we use the following preferences over consumption and labor hours from Greenwood et al. (1988):

$$u(c, \ell; \psi_i) = \frac{1}{1-\gamma} \left(c - \frac{\psi_i}{1+1/\eta} \ell^{1+1/\eta} \right)^{1-\gamma},$$

where c is the household's real annual equivalized consumption and ℓ is the annual labor hours of the household head or spouse. We will use the same preferences in our welfare analysis (Section 4) and discuss them further there. We express our results as a money-metric by rescaling each marginal utility using the population-average level.*

This carefully constructed regression yields coefficients which are exactly the object in Equation 3, that is, the left-hand side of the Baily–Chetty equation for the optimal level of social insurance (Baily, 1978; Chetty, 2006). We thus capture the social benefits of self-targeting, which Section 4 compares to calibrated estimates of the social costs of take-up. Panel D shows the estimates with and without a sample limitation to the simulated-eligible.

Panel D shows that, in some but not all transfers, the marginal dollar of transfer to recipients is worth substantially more than the marginal dollar to eligible nonrecipients. For instance, society would be willing to pay \$1.96 in resources to make a nondistortionary transfer of \$1 from the average SNAP-eligible nonrecipient to the average SNAP recipient. We estimate this Baily--Chetty premium to be 1.32 on average over the eight transfers. It arises because, as shown in the other three panels, transfers tend to attract take-up by households with higher “need” than is indicated by their income and eligibility status alone. Indeed, in transfers where we found no advantageous self-targeting on consumption and lifetime income, we do not find such a premium.”

** We calibrate $\gamma = 3$ (capturing both individual risk aversion and social welfare weights) and $\eta = 0.3$ as constants across households. The disutility parameter is calibrated internally as $\psi_{it} = w_{it} / \ell_{it}^{1/\eta}$, which we obtain by inverting the household's labor supply function $\ell_{it}^* = \ell_{it}^*(w_{it}, \psi_{it})$. Following Finkelstein et al. (2019), we also impose a “consumption floor” at the fifth percentile in calculating marginal utility.*

[Appears as footnote in paper.]

3) Section 3.1 raises the possibility of variation in preferences such that the mapping from consumption to marginal utility is not simple and monotone. We also tackle this empirically, by considering specific departures from our baseline model and asking if we would still find self-targeting in such cases. Essentially, our approach is to show that, within categories of households that are plausibly distinct in their preferences, we still see ample self-targeting into transfers. These results are all reported on p.18 in the main text.

- a) *Lifecycle changes in preferences.* Appendix Figure A9 (p.50) estimates self-targeting by age group, and Appendix Figure A8 (p.51) allows the control for income to have a cohort-specific slope (so that the self-targeting coefficient can be interpreted as an average of within-cohort comparisons). These figures adopt the recommendation of Blundell and Preston (1998) for within-cohort comparisons of consumption when assessing welfare.
- b) *Household composition affects utility function (children and marriage).* Appendix Figure A6 (p.48) allows for specific slopes in income by types of “household composition” (i.e., number of adults X number of children under 18 X marital status). This figure establishes that, even if these households have different marginal utility functions $u'(c, \theta)$, we would still find self-targeting on MU within these household types on average.
- c) *State-dependence in health and consumption–leisure complementarities.* Appendix Figure A10 (p. 52) estimates self-targeting by health status (poor or fair, good, very good or excellent) by and labor hours (nonemployed, part time, full time). We find self-targeting across levels of health status and levels of labor hours. As a result, we don’t need to take a stand on whether such state dependence / nonseparability exists; our claims of self-targeting on “need” (marginal utility) would go through anyway.

4) A portion of Section 3.3, titled “Consumption as a Measure of Need.” (p.17) also engages more on the Reviewer’s point that consumption may not be a good measure of household need, if it is highly endogenous to the tax-and-transfer system. Therefore, we were motivated to look for self-targeting on essentially exogenous predictors of need, and to bound how much endogeneity of consumption could matter. The Reviewer’s comments led us to add two new robustness-check analyses to the appendix:

- Appendix Table A4 (p.79, “Self-Targeting on Additional Measures of Need”) shows that using plausibly exogenous “tags” of need, we continue to find advantageous self-targeting into transfers. This evidence is persuasive if one sees observables like educational attainment as indicating need but being less elastic to taxes and transfers.
- Appendix Figure A4 (p.46, “Bounds Analysis of Self-Targeting on Consumption”) provides a bounds analysis showing that, unless behavioral responses to transfers are extraordinarily large (i.e., implausible MPCs), transfers would still be self-targeted on household consumption in a no-transfer counterfactual.

We quote the new text in full, which summarizes two new analyses in the appendix:

“We have so far described our results in ways that imply take-up is a ‘tag’ of some exogenous concept of need. Behavioral responses to taxes and transfers, however, suggest that household need is not fully exogenous. Three pieces of evidence suggest take-up does indeed reveal information about households’ exogenous resources.

*First, we find self-targeting in the most plausibly exogenous correlates of need (see **Appendix Table A4**). In most programs, take-up among the eligible is strongly predictive of high-school dropout, single parenthood, disability, minority race and ethnicity, and lower rates of savings. If these groups are high-need, then these patterns also provide support for the assumption that low consumption reveals marginal utility.*

[...]

*Third, we show that the behavioral responses in consumption required to fully explain our results are very large (see **Appendix Figure A4**). This responds to the concern that recipients adjust reported or actual consumption in response to transfers. For most transfers, the required response exceeds one dollar for every dollar of transfer received. In addition, our focus on the targeting of the ‘last dollar’ of transfers is conservative. Receiving transfers itself enables households to raise consumption and thus lowers imputed marginal utility, and so the ‘first-dollar’ targeting of transfers would thus be even better than in our main results.”*

Our full responses to their Comments 5 and 6 discuss a number of other edits. We hope our edits adequately address the Reviewer’s important questions on this matter. We think their suggestions truly helped to strengthen the manuscript, in particular leading us to provide discussion and analysis that bridges Section 3’s descriptive empirics and Section 4’s welfare analysis.

We wish to emphasize our willingness to undertake further revision of the paper on this point, if given further direction from the Editor and Reviewers.

2) “I also share the referee’s surprise ‘that a paper focused on the differences in income and consumption in the low income population has no discussion of the enormous literature on consumption smoothing, liquidity constraints, precautionary saving, insurance value of transfers, etc.’”

We thank the Editor for raising this point. We have made several major revisions to the paper and its analysis in light of the Reviewer’s comments on this point.

In summary, we made two major edits (corresponding to the same numbers immediately below):

- 1) The paper’s introduction now engages more fully with the literature on the role of the safety net in consumption smoothing (on p.2, p.5).

- 2) Section 3.3 contains a new subpart that seeks to connect our results with theories of consumption behavior (p.19).

In what follows, we quote from our responses to their Comments 4 and 6, editing our responses modestly to unify the presentation:

- 1) Engaging with the literature on consumption smoothing:
 - a) We now cite the two specific papers the Reviewer mentions, along with others we thought were especially related, in the introduction (p.5): *“The first literature often views the safety net as a form of insurance against consumption fluctuations (e.g., Blundell and Preston, 1998; Gruber, 2000; Blundell and Pistaferri, 2003; Blundell et al., 2008; Hoynes and Luttmer, 2011). In estimating the extent of consumption insurance from SNAP, for instance, Blundell and Pistaferri (2003) suggest that take-up is strongly influenced by ‘long-run income innovations,’ as we find.*
 - b) We also revised the introduction to point to consumption-smoothing and partial insurance as reasons why there are differences between income and consumption among the low-income population (p.2): *“Empirical research has consistently confirmed this premise (Poterba 1989, 1991; Cutler and Katz, 1992; Blundell and Preston, 1998), including for low-income households.”*
- 2) Following the Reviewer’s guidance, we also engage substantively with the connection between self-targeting and theories of consumption behavior (p.19).

“Theories of consumption behavior suggest several potential microfoundations for self-targeting. In particular, take-up might be driven by differences in expected future consumption, future-consumption risk, borrowing constraints, and tastes. Here we explore which of these appears important, in the hope of offering some directions for future research. Differences in two-years-ahead future consumption account for much of self-targeting (see Appendix Figure A19). That is, among households with similar current income and future consumption, take-up becomes much less informative of current consumption. By contrast, additional controls seem to matter little, including the household’s stock of liquid assets or the occupation and industry of household members. These findings provide little indication that borrowing constraints or precautionary-savings motives are important reasons for take-up. Similar to Blundell et al. (2008), we conclude that a simple life-cycle model with permanent and transitory components of income is likely to generate realistic self-targeting on consumption.”

Please see our reply to Reviewer 2’s Comment 6 for a full unpacking of this analysis.

Reviewer 1

We thank the referee for their careful review of our work.

1. “I think it would be valuable to at least discuss more extensively (and possibly briefly analyze) the considerable heterogeneity across programs. Is there any at least suggestive evidence on plausible mechanisms underlying the heterogeneity? That could be useful for drawing lessons that apply more broadly across multiple programs and for guiding future work on these programs.”

We thank the Reviewer for this valuable suggestion. With it in mind, we added two new exhibits in the Appendix which are summarized in the main text of the paper:

- Appendix Figure A17 (p.59) correlates the self-targeting coefficients with program characteristics. Our main finding is that ordeals and eligibility rules appear to act like substitutes, in that programs with especially stringent eligibility rules (like SSI) have weaker self-targeting than programs with less stringent rules (like SNAP).
- Appendix Figure A18 (p.60) shows that across-program heterogeneity is not “spurious,” i.e. arising from variation in the pool of eligible households across programs.
- Appendix Figure A1 (p.43) shows that self-targeting is essentially absent in contributory social programs like Social Security, Unemployment Insurance, and Worker’s Compensation.

These figures are now discussed in a subsection on across-program heterogeneity (p.19) For easy reference, we quote the across-program text (which is entirely new) in full below:

We provide a preliminary analysis of the predictors of the intensity of self-targeting across transfers. In particular, we correlate our estimates with the following program characteristics: the eligible share of the population, and the mean consumption and income ranks of the eligible. There is some suggestive evidence for the view that ordeals and eligibility rules act like substitutes: Programs with tighter eligibility rules have weaker self-targeting (see Appendix Figure A17). However, this pattern is not causal and may reflect other program attributes.¹

One further challenge in studying across-transfer heterogeneity is that it could arise spuriously from differences in eligibility (Appendix Figure A18).² To test this explanation, we build a “stacked” dataset with one observation for each person and each transfer for which that person

¹ Appendix Figure A1 shows that we do not find self-targeting in “contributory” social insurance programs like unemployment insurance, worker's compensation, and Social Security. Self-targeting on need would not be expected in these programs, as benefit levels are functions of past earnings. [Footnote appears in the paper text.]

² By “spurious,” we mean the following scenario. Every person is equally likely to take up Transfer A as Transfer B, but they vary in their probability of taking up either one. If different populations were eligible for A and for B, we may find different intensities of self-targeting in A and B. [Footnote appears in the paper text.]

is eligible. We then estimate models of take-up with and without person-level fixed effects.³ These specifications yield similar results, implying across-transfer heterogeneity in self-targeting is not spurious (i.e., is genuinely within-person).

The above text tries to avoid overclaiming about how much can be said about this heterogeneity, while providing some direction for future research. We hope it strikes the right balance. If the Reviewer desires any specific further effort to explain this across-transfer heterogeneity, please let us know.

2. “I like the welfare analysis and think the counterfactual policy you consider does a good job of isolating the core question of the paper. But it could be valuable to report other welfare-related results as well. For example, with similar assumptions to those you already make plus some outside evidence from the growing library of estimates of MVPFs (such as those in Hendren and Sprung-Keyser, 2020), it seems like you could calculate estimates of the MVPF of expanding the programs as they currently exist and of the MVPF of expanding the hypothetical automatic versions you consider. These could be ex ante MVPFs that account for within-lifetime insurance at least (if not across-type redistribution as well). (Sorry if I am missing something about why this would be difficult.) The contrast between the MVPFs of each actual vs. automatic program would be another way to make your main point about the value of self-targeting. And the estimates of MVPFs of the actual programs -- at least some of which, as far as I know, do not exist in the literature, at least not the ex ante version that reflects insurance value -- would help inform decisions of how to allocate government funds across programs.”

We thank the Reviewer for this useful suggestion. The Appendix has two new additions:

- A theoretical proposition deriving Marginal Values of Public Funds (MVPFs) of marginal increases in automatic transfers and voluntary transfers. See pp.113-115.
- An empirical calibration of the automatic-transfer and voluntary-transfer MVPFs for all eight programs in our study. See Appendix Table A17 (p.92), with associated discussion in Appendix C (p.107).

We have elected to put the MVPF analysis in the appendix, as its results echo the main welfare analysis.. We would be happy to move material from this analysis into the main text, if the Editor and Referee request it.

3. I think the paper would benefit from being clearer about the role of the distributional analysis of lifetime gains (Section 4). I found that interesting and useful, but my feeling is that the current draft isn't very clear about its value added relative to the correlation tests and the welfare analysis.

We thank the Reviewer for raising this important point. Following their recommendations, we have cut the distributional analysis entirely from the paper, with the goal of making the paper as tight and linear as possible. We are of course happy to undo this change, if requested by the Editor or Reviewers. We

³ The person-effects specification is identified only by people who are simultaneously eligible for multiple transfers, analogous to movers identifying firm or place effects. [Footnote appears in the paper text.]

originally saw the value-add of the distributional analysis as illustrating how self-targeting has important implications for standard policy analysis of the safety net. However, we agree that it is a less central contribution in the revised version.

4. “I agree that optimization failures that lead people to be less likely to take-up than is privately optimal imply smaller ordeal costs than the rational model. But optimization failures that lead people to be more likely to take up than is privately-optimal (as in Chan 2017, for example) could imply larger ordeal costs than the rational model. I'm not sure if there's anything very compelling you could do on this, but it seems worth at least raising as a possibility.”

We thank the Reviewer for this very nice suggestion. We have added a footnote on p.31, where we discuss the implications of optimization failures, to recognize the “two-sided” implications:

Other behavioral forces could raise take-up, such as limited self-control (Chan, 2017). If larger in magnitude than misperceptions, these would lead us to understate ordeal costs.

5. Page 1: I wasn't sure what you mean by "economists lack a way to connect empirical estimates of self-targeting to theory."

We thank the Reviewer for flagging this phrase as opaque. We have replaced it with the following text on p.1, which we hope is clearer:

Whether, in actual transfers, the benefits of self-targeting offset ordeal costs thus requires both new theory and careful measurement.

If the Reviewer would prefer another phrasing of this point, we welcome their suggestions!

6. It seems worth a brief explanation of why you focus on these eight programs. It sounds like it's mainly because those are what's available in the PSID, which is fine, but it's worth explaining what that captures vs. leaves out.

We thank the Reviewer for this point, and we made several edits to the “Transfer Receipt” portion of Section 2.1 (p.6) with it in mind.

- 1) We revised the main text of the paper to be more explicit that this is a data limitation: “*The PSID records self-reported household-level receipt for ten means-tested transfers.*” This text replaces language which had been less clear on this matter. Previously, this sentence was “*We observe self-reported household-level receipt for ten transfers.*”

Please note that, in presenting our analysis, we combine Section 8 and public housing (as housing assistance), along with school lunch and school breakfast (as school meals), yielding eight programs. The PSID includes all major federal transfer programs. However, it excludes important sources of income for the poor including tax credits like the EITC and Child Tax Credit. It also

excludes public education programs like Head Start, as well as state-specific programs. (In the quoted footnote below, we note these caveats.)

- 2) A new footnote also acknowledges that the PSID contains data on contributory (i.e., not means-tested) transfers. In particular, the footnote says in full (p.6):

“Our analysis excludes contributory social-insurance programs such as unemployment insurance, worker’s compensation, and Social Security. These programs differ fundamentally from noncontributory transfers, and accordingly, we see differences in self-targeting (see Appendix Figure A1). No other transfers are consistently available in the PSID, with the main omissions being tax credits and federal education subsidies.”

- 3) To emphasize, the footnote now refers to additional analysis on the contributory transfer programs that fall outside of the scope of our study, in Appendix Figure A1 (p.43). The key finding of that figure is that there is no self-targeting in these contributory programs.

We had excluded contributory programs (UI, Worker’s Comp, Social Security) from our analysis due to consistent feedback from other economists, who disagreed these programs were transfers akin to SNAP/Medicaid. These economists pointed to contributory programs’ “pay in, pay out” structure that made benefits a function of earnings history. Their benefit formulas therefore intentionally discourage take-up by households with low consumption and lifetime income. For this reason, prior theoretical analyses of self-targeting (e.g., Nichols-Zeckhauser) usually have programs like SNAP or Medicaid in mind, rather than UI. We can see a case that this core distinction between contributory and non-contributory social insurance actually advances our main results. It could be given more attention in the main text of the paper, if the Editor and Referees would like.

7. Page 31 on the calibration says that you take the value of the transfer from the PSID. What about for in-kind transfers? It seems best to use estimates of the value of those transfers to households rather than of the cost to the government.

We thank the Reviewer for their question. Section 2.2 (p.8) explains what we currently do:

“We augment PSID/CEX data on the dollar values of transfers with values from the Supplemental Poverty Measure module of the U.S. Current Population Survey (CPS). For SNAP, TANF, SSI, and LIHEAP, the PSID records the nominal value of transfers over various time periods, which we rebase as the per-capita annualized amount in 2020 constant dollars. The PSID does not include cash-equivalent values for in-kind transfers, namely Medicaid, Section 8, public housing, and WIC. We impute these amounts with the average values by household size and year reported in the CPS for all but WIC, where we use the national average benefit. The CPS values in-kind transfers other than Medicaid dollar-for-dollar with expenditures.”

The alternative to the dollar-for-dollar assumption is to rescale in-kind transfers by some estimate of household WTP per in-kind transfer dollar. This would have no impact on Figure 1, and it would rescale

Figures 2 and 3 by a constant. We also explain the impact of such a change on Table 3 in the main text (p.31):

“Fourth, our results extend immediately to in-kind transfers, or other transfers that households may value differently from their cost of provision. Suppose households are willing to pay $\lambda < 1$ for each transfer dollar. Our expression for dW/ds from Proposition 1 would then be multiplied by λ , as we show in Appendix D. The welfare effect dW/ds remains correct in units of households' willingness-to-pay for the transfer, rather than in dollars. The parameter λ simply rescales the welfare effect and cannot reverse its sign.”

We have made this choice for three reasons:

- 1) Valuing transfers at their cost of provision follows U.S. official statistics on poverty measurement (like the Supplemental Poverty Measure). It is also common in descriptive papers in the literature: the assumption is carried by both Auten and Splinter (2024) and Piketty, Saez, and Zucman (2018), as well as Blanchet, Chancel, and Gethin (2022) and Gethin (2024), among others.
- 2) Choosing the benchmark of valuing transfers dollar-for-dollar is transparent and a natural benchmark, since many programs cover (what should be) inframarginal consumption (e.g., SNAP and school lunch). Whether transfers are more or less valuable than their cost of provision is not clear ex-ante, nor is there a scholarly consensus on such questions.⁴ Taking a stance on the correct discount or premium would thus be controversial and would require models of WTP for each program. Then, differences across programs would be difficult to evaluate, as it would not be clear whether they stem from model misspecification.
- 3) Critically, we construct our reform to be robust to this concern. As discussed in the passage above, the sign of our welfare conclusions are insensitive to whether transfers are valued at face value or not. This is a virtue of our analysis.

8. Page 35: I believe the following should read, "[...] automating transfers is likely to be desirable only when people care *less* about consumption fluctuations [...]"

Thanks for catching this typo. Fixed!

9. Appendix Figure A8: This figure would be more useful if the data labels did not overlap so much.

Thanks so much for this suggestion. We have revised the figure accordingly.

⁴ There could also be social externalities from providing food, for instance, if it increases health or malnourishment.

Reviewer 2

We thank the referee for their thoughtful suggestions and careful engagement with our paper.

1. “But a first question: why are the authors using the PSID and not the Consumer Expenditure Survey (CE)? The CE is universally used in the US for the calculation of consumption. It asks very detailed questions on consumption, far more than the PSID, is available for many years, and does not have the sample size problems that the PSID does. The PSID is subject to unknown attrition bias, having suffered huge attrition in its first generation—those alive when it started in 1968—with consequently unknown attrition bias in the second and third generations that are now part of the sample. Yes, the paper wants to do lifetime income, and panel data are needed for that, but that could be a supplementary analysis. The PSID has slightly better data on transfer program receipt than the CE, but it’s unclear if that outweighs the CE’s advantages.”

We thank the Reviewer for raising this important point. We aim to provide two main responses in what follows. First, we carefully replicate our PSID analysis in the CEX. Second, we then discuss the value-added of the PSID relative to the CEX (i.e., why use the PSID at all?).

CEX replication. We now present CEX results alongside PSID results throughout the main text of the paper. This was a substantial undertaking: We carefully built variables in the CEX that align with our PSID concepts (including the measure of consumption that adjusts for durable-goods outlays and all the eligibility simulations), and we went exhibit-by-exhibit through the paper, even replicating the PSID robustness checks in the CEX.

Our main takeaway is that our analysis fully replicates in the CEX. That is, there is no analysis that we did in the PSID that “breaks” when we redo it in the CEX. Once again, we thank the referee for urging us to do this replication exercise, and we hope it substantially strengthens the paper.

We review all exhibits and their discussion in the text:

- Figure 2 (p.21), “Estimates of Self-Targeting by Data Source (PSID Versus CEX)”

“We estimate self-targeting on consumption for the five transfers that appear in both the PSID and CEX. Overall, the CEX shows a similar degree of self-targeting as the PSID (see Figure 2). However, we find less heterogeneity across programs in the CEX than in the PSID. These results address the possibility that limitations that are specific the PSID, such as panel attrition or less-thorough survey questions on consumption, could explain our results.”

- Figure A14 (p.56), “Distributional Decompositions of Self-Targeting by Consumption Ventile”

This figure has a new Panel C, which shows that for SNAP and Medicaid in the CEX, take-up is particularly informative of having a very low consumption rank. Panels A and B, unchanged from the prior draft, showed this fact in the PSID with respect to both consumption and lifetime income

(also for SNAP and Medicaid). Accordingly, the main text now introduces this figure with the following text (p.19):

“Finally, we look more granularly at where self-targeting emerges across the distributions of consumption, lifetime income, and current income in the PSID and CEX. Households with the lowest consumption and lifetime income drive our self-targeting estimates (Appendix Figure A14).”

Previously, the original-draft text read on p. 14: *“Appendix Figure A5 shows our self-targeting results are driven by households with very low consumption and lifetime income.”*

- Figure A16 (p.58), “Heterogeneous Self-Targeting on Consumption by Current Income Rank (CEX)”

This figure is the CEX analog of Figure A15 for the PSID. Both figures are noted in the main text as follows (on p.19): *“Self-targeting on consumption also appears slightly stronger at the bottom of the income distribution than elsewhere (see Appendix Figures A15 and A16).”*

Previously, this text said (on p.14): *“Appendix Figure A7 shows a roughly similar extent on self-targeting on consumption and lifetime income above and below the tenth percentile of current-income rank.”* We have revised our reading of the results because the PSID analysis showed modest (insignificant) heterogeneity, whereas the CEX results are more precise.

- Figure A12 (p.54), “Demographic Heterogeneity in Self-Targeting (CEX)”

This figure is the CEX analog of Appendix Figure A11 for the PSID. Both figures are noted in p.18 of the paper, where we say the following: *“We first extend our analysis by dividing the population by education, race, and ethnicity (Appendix Figures A11 and A12). We also break out three disadvantaged populations: single parents, non-native English speakers, and car non-owners. Examining SNAP and Medicaid, we find some suggestive heterogeneity in self-targeting. In general, we see more self-targeting among better-off groups (e.g., college graduates) than worse-off groups (e.g., single parents). No single demographic category, however, drives our results or stands out strongly from this broad pattern.”*

The original discussion previously appeared on p.16 and read as follows: *“Finally, we examine whether self-targeting may be less informative of need for members of disadvantaged demographic groups. Such heterogeneity is motivated by the concern that ordeals screen out the most vulnerable groups (Mullainathan and Shafir, 2013), often reflected in the existence of targeted policy accommodations (such as social workers and native-language forms). We divide the population by education and race/ethnicity, and we also break out populations of particular interest, such as non-native English speakers and car non-owners. Appendix Figure A8 shows some suggestive heterogeneity in self-targeting, focusing on SNAP and Medicaid. In SNAP, for example, self-targeting appears to be weaker for Hispanics and non-native English speakers in SNAP, although these patterns are not so evident in Medicaid.”*

We have cut the mention of the suggestive patterns in the PSID about Hispanic ethnicity because this pattern is not apparent in the CEX. Since it only appeared in SNAP and not Medicaid in the PSID, we had treated this finding cautiously before in any case.

- Figure A13 (p.55), “Self-Targeting by Number of Distinct Transfers Received”

This figure has been updated to include CEX estimates of self-targeting by the number of distinct transfers received. Our estimates are almost indistinguishable from the PSID estimates (for consumption). In the paper, we now say (p.18): *“In both the PSID and the CEX, eligible households that receive multiple transfers are more self-targeted on consumption and lifetime income than those receiving only one transfer.”* This result was previously presented as part of Fact 4, so the discussion has been somewhat compressed from the original manuscript.

- Figure A21 (p.63), “Adjustments to Eligibility Simulations”

This figure is the CEX analog of Appendix Figure A20 for the PSID. Both figures are in the main text at p.20, where we say: *“First, simple adjustments to the eligibility simulations, such as imposing income limits or liquid-asset tests, have modest impacts on our estimates of self-targeting (Appendix Figures A20 and A21). These results suggest that further improvements to the eligibility simulations are unlikely to change our estimates materially.”*

- Figure A24 (p.67), “Self-Targeting on Consumption Among the Very Likely Eligible (CEX)”

This figure is the CEX analog of Figure A23 for the PSID. We mention both figures in the paper at p.21, writing: *“Second, when we restrict the sample to demographic groups with a very high probability of eligibility, we continue to find self-targeting (Appendix Figures A23 and A24). Among such groups, patterns of transfer receipt are less likely to be confounded by eligibility mismeasurement and thus more credibly isolate self-targeting.”* This discussion is essentially unchanged from p.17 of the original manuscript.

- Table A2 (p.76), “SNAP Receipt, Eligibility, and Take-Up Rates by Income and Consumption Quintile: Consumer Expenditure Survey”

This table is the CEX analog of Table 2 for the PSID. We direct readers to Table A2 in the notes to Table 2 on p.11. It is not discussed elsewhere.

- Table A3 (p.77), “Dollar and Percentage Differences Between Recipients and Similar-Income Non-Recipients”

This table now presents regression estimates of self-targeting for the CEX alongside previously-reported PSID estimates. The content of these tables are percentage-difference and dollar-differences in consumption and lifetime income between transfer recipients and

simulated-eligible nonrecipients at the same income rank. The discussion in the paper text (at p.14) is unchanged from the original manuscript.

- Table A6 (p.81), “Selection into Transfers, Adjusted for Misreporting of Transfer Receipt”

This table previously used functions from Mittag (2019) and Davern et al. (2019) to correct for misreporting of transfer receipt in SNAP and Medicaid respectively. The table previously showed the results of these corrections in the PSID. We have revised it to also apply these adjustments to the CEX. We have not altered the discussion of this table on p.22 of the main text.

- Table A9 (p.84), “Sensitivity of Self-Targeting on Consumption to Income Mismeasurement (CEX)”

This table is the CEX analog of Table A8 for the PSID. Both Tables A8 and A9 are new. It shows how our estimates of self-targeting change when we constrain the coefficient on income rank to a higher level than the OLS estimate, as a way to compensate for attenuation bias. We quote the relevant discussion on p.23 in full, at Comment 4 raised by this Reviewer.

- Table A11 (p.86), “Well-Measured Consumption and Transfer Receipt (CEX)”

This table is the CEX analog of Table A10 for the PSID. We mention this table at p.23 of the paper, writing that: “*Meyer and Sullivan (2023) describe consumption as comparatively ‘well-measured’ for some goods and services. Appendix Tables A10 and A11 document self-targeting in the PSID and CEX when we look exclusively at these consumption categories.*” This discussion is minimally edited for clarity from the discussion in the original manuscript at p.18.

- Table A13 (p.88), “Durable-Goods Ownership and Transfer Receipt (CEX)”

This table is the CEX analog of Table A12 for the PSID. We mention this table in the paper at p.23, writing that: “*Similar patterns [of self-targeting] manifest in PSID and CEX measures of household ownership of consumer durable goods.*” The associated text is unchanged from the original manuscript.

Value-Added of the PSID. While the analysis replicates in the CEX, the paper does maintain a focus on the PSID data. The PSID has three advantages relative to the CEX, which informed this choice:

1. We (and Referee 3) see a key empirical threat to the paper as being measurement error in eligibility. The CEX has more limited demographic information for imputing eligibility than the PSID, because the survey questionnaire focuses on expenditure. Limited demographic detail can affect the accuracy of eligibility imputations and complicate our key test of self-targeting, to the extent it results in ineligible people included in our analysis.

2. The CEX's coverage of means-tested transfers is less extensive than that of the PSID. In particular, it is missing WIC, LIHEAP, and school meals. The breadth of our findings across transfers is, we think, a key strength of the paper.
3. The CEX naturally does not record income histories, because it is not a panel dataset. There are several parts of the paper that become impossible without these data.
 - a. It is challenging to estimate transfers' insurance value, which we show empirically to be of substantial importance to the welfare impacts of self-targeting. We see the decomposition of insurance versus redistribution as an important conceptual part of the paper. This also relates to the Referee's encouragement below to focus on the relationship between the paper and previous literature on the insurance value of transfers.
 - b. The comparison of lifetime income and consumption lets us understand that the observed self-targeting on consumption reflects self-targeting on lifetime income, rather than on uninsurable but temporary shocks.

We can certainly appreciate the Reviewer's desire to have this paper use the best-available data in its core analysis. What we wish to emphasize is that these strengths also come with limits, and that by presenting CEX and PSID analyses side-by-side, we hope the paper now makes the best of both datasets.

We have also revised Appendix B.1 (p.94) to include detail on our CEX data cleaning procedures, alongside previous discussions of the PSID.

2. "One odd feature of the analysis is the sole focus on selection on consumption rather than on Y. The government uses Y, and it positively selects on Y: receipt is lower, the higher the income. My impression is that this is the main measure of selection that is used in the literature: whether those with greater need, measured by pretax pretransfer income, are more likely to participate. Table 2 shows that this is already the case in the paper's data, so the paper is already finding positive selection on need. Yes, it is even more positively selected when looking at consumption conditional on income, so the paper is just finding that selection is more positive than we thought."

Thank you for raising this point, as we think it hits at a key conceptual thrust of the paper. We made several edits to the paper to address it.

1) *Theoretical rationale for focus on consumption.* In brief, taxable income is observable. To the extent that transfers redistribute it, they duplicate the tax system. The essential idea of self-targeting is that transfers might solve an information problem, helping to reveal "need" that the government does not already observe. The paper is therefore focused on the unique benefits that transfers might already achieve, ones that could not be shifted to taxes. In part for these reasons, other papers in this literature also use consumption as a proxy of need (e.g., Deshpande and Lockwood, 2022).

We revised the paper to clarify this point, at Section 2.3 (p.11) and Section 3.1 (p.13):

“Self-targeting is of particular interest, as then take-up may reveal private information and thus relax incentive constraints on redistribution... These patterns suggest that transfers may indeed reveal something about need that is not already contained in their income.” (p.11)

“[T]he inclusion of flexible controls for current income reflects the theoretical perspective that transfers are designed to complement, not duplicate, income taxation.” (p.13)

This theoretical perspective is why we do not entirely agree that we are “just finding that selection is more positive than we thought.” This is fundamentally a matter of whether the government can achieve a welfare gain, holding the income distribution fixed, through the use of nontax instruments.

2) *Practical rationale for focus on consumption.* Economists have generally argued that consumption is superior to income as a proxy of “need”: consumption-smoothing and income mismeasurement. We have revised the introduction (on p.2) to provide those rationales upfront:

“These selection measures follow in a tradition in public finance dating to Vickrey (1947), one that takes as a premise that consumption is a superior measure of living standards to current income, insofar as households are at least partially insured against transient income shocks. Empirical research has consistently confirmed this premise (Poterba 1989, 1991; Cutler and Katz, 1992; Blundell and Preston, 1998), including for low-income households. More recent work highlights income mismeasurement among low-income households as an additional reason to favor consumption as a welfare measure (Meyer and Sullivan, 2003; Brewer et al., 2017).”

While economists are longtime critics of distributional analysis by income, policymakers continue to use it almost exclusively (see, e.g., the U.S. Congressional Budget Office and Joint Committee on Taxation). To the best of our knowledge, our paper is the first to show that there is an enormous amount of “hidden progressivity” of transfers in consumption and lifetime income.

We have added a sentence to Section 3.2 (p.14) that shows that, even if we are just finding that selection is better than we thought, that this revision to our understanding is big:

“These differences are so large that they often exceed the raw differences in average income ranks between recipients and simulated-eligible nonrecipients (Appendix Figure A2).”

In this sense, the U.S. government is getting a quite misleading picture of transfer programs by taking the shortcut of income as a proxy for need. Our MVPF analysis also shows that the bulk of the welfare gains from transfers come from redistribution *within* rather than *across* income levels (see p.107).

We are very open to suggestions from the Reviewer as to how best we can make either case for our analysis, if they desire further revisions.

3. “But an immediate problem is that income is measured with a high degree of error--and greater error than consumption, as the authors correctly note the literature shows. So the lower selection on income than on consumption shown in Panel A could be entirely the result of measurement error in

income. Although one hypothesis for the difference in selection on income and consumption in Panel A is measurement error in income, it may be that the paper’s interpretation is right: that low income households do consumption smoothing (that is not directly stated in the paper but it seems to be the interpretation, especially in the use of lifetime income as a ranking variable).”

We thank the Reviewer for raising their concerns about measurement error in income. We have made several revisions to the paper to improve our treatment of this issue:

1. We are more explicit about the threat in Section 3.4 (p.22-23). After mentioning the literature’s view that measurement error is greater in income than in consumption, we then say: *“This point threatens our analysis: To the extent we under-control for income due to attenuation bias, our estimates of self-targeting may be overstated.”*
2. We provide a new sensitivity analysis to income mismeasurement on p.23, which we quote in full here. The key idea of this check is to use the rank-rank relationship between income and consumption to have an upper bound on the plausible degree of mismeasurement in income. We think this check zooms in directly on the Reviewer’s concern and in particular shows that self-targeting is robust to any plausible degree of income mismeasurement:

Next, we directly probe the sensitivity of self-targeting to any bias transmitted from income mismeasurement. In a linear version of Equation 4, we constrain the coefficient on income rank to higher values than the unconstrained estimate. We claim there exists a threshold value for the constrained coefficient that is implausible a priori, as it would imply an excessive noise share of income variance or unrealistically little smoothing of consumption relative to income. See Appendix C for further explanation of this procedure.*

Self-targeting in most transfers appears robust to substantial mismeasurement of income (see Appendix Tables A9, A10, and A11). In the PSID, when we constrain the rank–rank slope between consumption and income to 0.8, up from 0.51 when unconstrained, we find that SNAP recipients are on average 12.4 percentiles lower in the consumption distribution than eligible nonrecipients with the same income. This amounts to a downward adjustment of only 2.1 percentiles from our baseline estimate, despite the assumption of a substantial attenuation bias in the rank–rank slope.

[In footnote: $\bar{R}_i = \beta D_{it} + \gamma R_{it} + u_i$ where $\bar{R}_i$ is consumption rank or lifetime rank, R_{it} is income rank, and D_{it} is transfer receipt.]*

To provide a bit more econometric context on this sensitivity analysis: It reuses the idea of Anderson-Rubin IV confidence intervals. Anderson-Rubin intervals are constructed by finding the interval of β at which the coefficient on γ is zero: $y_i - \beta x_i = \gamma z_i + e_i$.

This should already look like our setup: We're constructing the interval of β (coefficient on income rank) consistent with $\gamma = 0$ (moment condition, that transfer receipt has no direct effect). To see the connection with our work, suppose transfer receipt would not directly predict consumption rank given a perfectly-measured income rank. Then receipt is a valid instrument for the mismeasured income rank. Thus, if there is no sensible coefficient on income rank at which the coefficient on transfer receipt = 0, then transfer receipt cannot be an instrument. By implication, our finding of self-targeting couldn't be explained by income mismeasurement.

We wish to remark here on a potential further revision we could undertake to address measurement error, if the Editor and Reviewers advise:

- We could augment the model of Blundell et al. (2008) with a parametric model of the transfer take-up decision: $\Pr(D_{i,a,t} = 1 | c_{i,a,t}, y_{i,a,t}, Z_{i,a,t}) = \Phi(c_{i,a,t}, y_{i,a,t}, Z_{i,a,t}; \theta)$, where $c_{i,a,t}$ is true consumption (free of measurement error), $y_{i,a,t}$ is true income (free of measurement error), and $Z_{i,a,t}$ are observable characteristics affecting tastes and preferences. Let θ be a vector of parameters governing the take-up decision.
- Suppose we first re-estimate their model on our sample, we then simulate the income/consumption data. And also suppose we initialize some take-up equation parameters θ . Using method of simulated moments, we could find the θ such that we minimized a distance between simulated and actual conditional take-up probabilities $\Pr(D_{i,a,t} = 1 | \hat{c}_{i,a,t}, \hat{y}_{i,a,t}, Z_{i,a,t})$, using hats to the conditional probabilities are computed using observed consumption and observed income (i.e., with measurement error).

4. “It is rather surprising that a paper focused on the differences in income and consumption in the low income population has no discussion of the enormous literature on consumption smoothing, liquidity constraints, precautionary saving, insurance value of transfers, etc...Likewise, on the insurance side, low-income families have alternative sources of insurance. There is a large, unreferenced literature on the insurance value of transfers (examples are Blundell et al., AER, 2008; Blundell-Pistaferri, JHR, 2003). Implicit in that literature is that transfers may not have full insurance value.”

We thank the Reviewer for this encouragement to engage more with the literature on the role of the safety net in consumption smoothing. We have taken the following actions in response to their points:

1. We now cite the two specific papers the Reviewer mentions, along with others we thought were especially related, in the introduction (p.5): *“The first literature often views the safety net as a form of insurance against consumption fluctuations (e.g., Blundell and Preston, 1998; Gruber, 2000; Blundell and Pistaferri, 2003; Blundell et al., 2008; Hoynes and Luttmer, 2011). In estimating the extent of consumption insurance from SNAP, for instance, Blundell and Pistaferri (2003) suggest that take-up is strongly influenced by ‘long-run income innovations,’ as we find.*

2. We also revised the introduction to point to consumption-smoothing and partial insurance as reasons why there are differences between income and consumption among the low-income population (p.2): *“Empirical research has consistently confirmed this premise (Poterba 1989, 1991; Cutler and Katz, 1992; Blundell and Preston, 1998), including for low-income households.”*

5. “There are many researchers who work on the low-income population who think that low-income households are completely liquidity constrained--they cannot borrow, they do not save, etc. If that were the case, income in a period (properly measured) would equal consumption in the period. The data here reject that, but it could be the result of measurement error in income. But the paper rejects that hypothesis in its analysis of measurement error in income--it takes the difference as true (see below for that analysis).”

We appreciate the Reviewer’s comments that measurement error in income, and its particular connection to savings-like behaviors. We engage with their concern about income mismeasurement in reply to their Comment 4.

6. “But take, first, the literature on precautionary saving (Hubbard-Skinner-Zeldes JPE 1995 is the seminal reference). Transfer programs reduce that saving. Further, there is heterogeneity across households in the degree to which they do that. Those who do not save will have less liquid assets in the future and will rely more on transfer programs. So those in the bottom quintile of consumption in the receipt of programs may be those who didn’t save.”

We have embraced the Reviewer’s suggestion to engage more with the precautionary savings literature. To that end, Section 3.3 contains a new subpart that seeks to connect our results with theories of consumption behavior (p.19). It reads in full:

“Theories of consumption behavior suggest several potential microfoundations for self-targeting. In particular, take-up might be driven by differences in expected future consumption, future-consumption risk, borrowing constraints, and tastes. Here we explore which of these appears important, in the hope of offering some directions for future research. Differences in two-years-ahead future consumption account for much of self-targeting (see Appendix Figure A26). That is, among households with similar current income and future consumption, take-up becomes much less informative of current consumption. By contrast, additional controls seem to matter little, including the household’s stock of liquid assets or the occupation and industry of household members. These findings provide little indication that borrowing constraints or precautionary-savings motives are important reasons for take-up. Similar to Blundell et al. (2008), we conclude that a simple life-cycle model with permanent and transitory components of income is likely to generate realistic self-targeting on consumption.”

To explain further, the analysis includes a new appendix figure (A19, on p.61). This figure explores four theories for why we find self-targeting, here explained in greater detail.

- a) **Consumption smoothing (LC PIH):** Consider a consumption-savings model without precautionary saving or borrowing. Transfer recipients take up because they have low income or anticipate low future income. Then differences in *present* consumption across recipients and nonrecipients should vanish when we compare groups with the same *future* consumption. We do this using the panel structure of the PSID, and we find it does indeed account for much of self-targeting.
- b) **Borrowing constraints:** If take-up is providing partial consumption insurance that is inferior to dissaving, then households should do it when they are constrained. We show in the figure that wealth differences explain little of the heterogeneity in present consumption given income, after accounting for future consumption.
- c) **Income risk (precautionary saving).** After accounting for current income, future consumption, and wealth, differences in current consumption between transfer recipients and simulated-eligible nonrecipients are small. When we try to also account for education/industry/occupation, as proxies for households' degree of income risk, we find we can eliminate little of the remaining difference in consumption ranks.
- d) **Tastes (e.g., differences in discount rates).** A theoretical interpretation of the residual differences in consumption rank would look to tastes, although we do not distinguish the component of this which is measurement error.

Chetty (2006) explains why our measure of the welfare benefits of self-targeting would not be affected by the size of a precautionary savings response to transfer programs. Transfers might, through incentives (not) to save, affect the marginal utilities of income among transfer recipients and nonrecipients. But, within some class of models, the difference in marginal utilities would be a sufficient statistic for welfare benefits.⁵

The welfare-relevance of the savings response would run through its fiscal externality, for instance, more people might satisfy an asset test and thus get benefits. (This is how we see the potential connection to Hubbard et al. 1995.) While we don't currently account for fiscal externalities from saving, we could mention this point in the paper or attempt an estimate, if the Reviewer thinks we should.

7. "In both of these cases, the problem is that one needs a counterfactual: how would consumption today (and last year?) be different in the absence of transfers? One needs a behavioral model for that."

We thank the Reviewer for this helpful comment. We agree, with the minor reframing we need the model not to construct counterfactual consumption but rather to get marginal utilities of consumption. Stepping back, if one wants to ask if transfers are well-targeted on "need" (that is, on marginal utility), one inescapably needs a utility model that yields some mapping $u'(c; \theta)$ from consumption to marginal utility.

⁵ Chetty, R., 2006. A general formula for the optimal level of social insurance. *Journal of Public Economics*, 90(10-11), pp.1879-1901.

We have made substantial changes to the paper, both in the form of expositional changes and new analysis. We summarize the changes made:

1. Section 3.1 (pp.11-13) has been completely rewritten, to address this comment and others made by Reviewers 2 and 3. We now define, explicitly and upfront, the theoretical object of ultimate interest: $[E[u_{c,1}'] - E[u_{c,0}']] / E[u_c']$, which we explain as follows:

“The ideal measure of self-targeting in a given transfer is the average difference in the social marginal utilities of income (“need”) between the transfer’s recipients and eligible nonrecipients who are otherwise identical to the government...where $u_{c,1}'$ and $u_{c,0}'$ respectively indicate the marginal utilities of recipients and eligible nonrecipients. The denominator, average marginal utility, expresses this difference in money-metric terms. That is, the measure in Equation 3 captures a society’s estimated willingness to pay (per dollar) to redistribute resources from eligible non-recipients to recipients, under the assumption that fiscal externalities from this action are offset via lump-sum taxes on all eligible individuals.”

We then emphasize the inescapable role of a model in this section:

“First, marginal utilities of income are unobserved, requiring a model to link the data to marginal utility. Here, we take consumption as a proxy for marginal utility, as others have commonly assumed (e.g., Finkelstein et al., 2019; Deshpande and Lockwood, 2022). Two broad motivations for this choice are the partial insurance of household consumption against income shocks and the mismeasurement of low incomes in both survey and administrative data. This choice aligns with consumption–savings models where marginal utility is monotonic in consumption. Under this monotonicity, consumption ranks align with marginal-utility ranks in a cross-section of households. Lifetime-income ranks would also align in this sense if consumption is fully insured against income shocks. We therefore describe transfers as “advantageously self-targeted” if take-up among eligible households predicts lower ranks in the consumption or lifetime-income distributions.”

2. We said above that we step more patiently from data description (rank-rank regressions) to welfare interpretation to make more clear the role of theory. We do this in our new “Fact 4” (pp.15-17). We quote this new text in full, and highlight the new figure (Panel D of Figure 1):

“Marginal utilities of income appear higher among transfer recipients than among eligible nonrecipients with similar incomes. Panel D connects our descriptive self-targeting regressions to the welfare analysis, and specifically to the welfare benefits of self-targeting. We do so by replacing the household-rank dependent variable with a calibrated value for each household’s marginal utility of income. The resulting coefficients can be interpreted as the additional value of moving a marginal dollar from a transfer’s eligible nonrecipients to its recipients. To calibrate marginal utilities for each

household, we use the following preferences over consumption and labor hours from Greenwood et al. (1988):

$$u(c, \ell; \psi_i) = \frac{1}{1-\gamma} \left(c - \frac{\psi_i}{1+1/\eta} \ell^{1+1/\eta} \right)^{1-\gamma},$$

where c is the household's real annual equivalized consumption and ℓ is the annual labor hours of the household head or spouse. We will use the same preferences in our welfare analysis (Section 4) and discuss them further there.* We express our results as a money-metric by rescaling each marginal utility using the population-average level.

This carefully constructed regression yields coefficients which are exactly the object in Equation 3, that is, the left-hand side of the Baily–Chetty equation for the optimal level of social insurance (Baily, 1978; Chetty, 2006). We thus capture the social benefits of self-targeting, which Section 4 compares to calibrated estimates of the social costs of take-up. Panel D shows the estimates with and without a sample limitation to the simulated-eligible.

Panel D shows that, in some but not all transfers, the marginal dollar of transfer to recipients is worth substantially more than the marginal dollar to eligible nonrecipients. For instance, society would be willing to pay \$1.96 in resources to make a nondistortionary transfer of \$1 from the average SNAP-eligible nonrecipient to the average SNAP recipient. We estimate this Baily--Chetty premium to be 1.32 on average over the eight transfers. It arises because, as shown in the other three panels, transfers tend to attract take-up by households with higher “need” than is indicated by their income and eligibility status alone. Indeed, in transfers where we found no advantageous self-targeting on consumption and lifetime income, we do not find such a premium.”

* We calibrate $\gamma = 3$ (capturing both individual risk aversion and social welfare weights) and $\eta = 0.3$ as constants across households. The disutility parameter is calibrated internally as $\psi_{it} = w_{it} / \ell_{it}^{1/\eta}$, which we obtain by inverting the household's labor supply function $\ell_{it}^* = \ell_{it}^*(w_{it}, \psi_{it})$. Following Finkelstein et al. (2019), we also impose a “consumption floor” at the fifth percentile in calculating marginal utility.
[Appears as footnote in paper.]

3. A portion of Section 3.3, titled “Consumption as a Measure of Need.” (p.17) also engages more on the Reviewer’s point that consumption may not be a good measure of household need, if it is highly endogenous to the tax-and-transfer system. Therefore, we were motivated to look for self-targeting on essentially exogenous predictors of need, and to bound how much endogeneity of consumption could matter. The Reviewer’s comments led us to add two new robustness-check analyses to the appendix:

- Appendix Table A4 (p.79, “Self-Targeting on Additional Measures of Need”) shows that using plausibly exogenous “tags” of need, we continue to find advantageous self-targeting into transfers. This evidence is persuasive if one sees observables like educational attainment as indicating need but being less elastic to taxes and transfers.
- Appendix Figure A4 (p.46, “Bounds Analysis of Self-Targeting on Consumption”) provides a bounds analysis showing that, unless behavioral responses to transfers are extraordinarily large (i.e., implausible MPCs), transfers would still be self-targeted on household consumption in a no-transfer counterfactual.

We quote the new text in full, which summarizes two new analyses in the appendix:

“We have so far described our results in ways that imply take-up is a ‘tag’ of some exogenous concept of need. Behavioral responses to taxes and transfers, however, suggest that household need is not fully exogenous. Three pieces of evidence suggest take-up does indeed reveal information about households’ exogenous resources.

*First, we find self-targeting in the most plausibly exogenous correlates of need (see **Appendix Table A4**). In most programs, take-up among the eligible is strongly predictive of high-school dropout, single parenthood, disability, minority race and ethnicity, and lower rates of savings. If these groups are high-need, then these patterns also provide support for the assumption that low consumption reveals marginal utility.*

[...]

*Third, we show that the behavioral responses in consumption required to fully explain our results are very large (see **Appendix Figure A4**). This responds to the concern that recipients adjust reported or actual consumption in response to transfers. For most transfers, the required response exceeds one dollar for every dollar of transfer received. In addition, our focus on the targeting of the ‘last dollar’ of transfers is conservative. Receiving transfers itself enables households to raise consumption and thus lowers imputed marginal utility, and so the ‘first-dollar’ targeting of transfers would thus be even better than in our main results.”*

4. The original manuscript also provided evidence that is strongly suggestive of self-targeting on need even if one has concerns about our main consumption outcome. First, we show substantial evidence of self-targeting on lifetime income; while lifetime income can of course also be debated as a measure of need, consistency across measures is reassuring. Second, even if one is skeptical of using an aggregate consumption measure to proxy for need, it is likely that some components of consumption do indeed reflect need. We examine such aspects on p.23–24, and usually find evidence of need on such dimensions. (On the latter, the CEX data that the Reviewer prefers are particularly consistent in Table A13.)

We hope these edits have captured the Reviewer’s point. We welcome any specific suggestions they have.

8. “And the larger problem is that consumption is not a very good measure of need—which this paper assumes it to be. It is endogenous and chosen by households with life cycle considerations and in light of what liquidity constraints they face. It is not a measure of resources; it is a measure of how households allocate their resources across different periods, according to their tastes and preferences. Most analysts of poverty think that a measure of exogenous resources should be used to measure need, not an endogenous variable like consumption.”

We thank the Reviewer for this comment, which motivated the substantial edits we quote above in our reply to their Comment 5. Our paper’s welfare analysis takes as a baseline that consumption ranks “align” with ranks of marginal utility. But the Reviewer’s concern is very reasonable, as clearly the mapping $u'(c, \theta)$ from consumption to marginal utility can be complex: not homogenous across the population, shifting due to lifecycle patterns, and potentially state dependent.

To address this comment, we test types of variation in preferences that might break the monotonic mapping from consumption to marginal utility. We structure our response by proposing specific departures from models in which marginal utility is monotonically decreasing in consumption; we surely do not cover every possible model, and if there are specific departures that the Reviewer would like to raise that we missed, we would welcome the opportunity to address them.

We review here several additions to the paper which address the Reviewer’s important point:

- 1) Section 3.1 (p.12), fully rewritten, now alerts readers to the precise objection the Reviewer raises about tastes.

“Nonetheless, imputing marginal utility from consumption has inherent limitations. Perhaps the most significant is that it is infeasible to fully capture all variation in preferences when performing this imputation. For instance, factors like aging, parenthood (Blundell and Preston, 1998), opportunity costs of time (Aguiar and Hurst, 2005), and habit formation (Chetty and Szeidl, 2016) all influence the relationship between consumption and marginal utility. Heterogeneity in labor-supply elasticities or nonseparability of labor and consumption create similar difficulties.”

- 2) We then show robustness to what we saw as the most natural forms of taste heterogeneity. Our approach is to show that, within categories of households that are plausibly distinct in their tastes, we still see ample self-targeting into transfers. These results are all reported on pp.23-24 in the main text.
 - d) Appendix Figure A9 (p.50) estimates self-targeting by age group, and Appendix Figure A8 (p.51) allows the control for income to have a cohort-specific slope (so that the self-targeting coefficient can be interpreted as an average of within-cohort comparisons). These figures adopt the recommendation of Blundell and Preston (1998) for within-cohort comparisons of consumption when assessing welfare.

- e) Appendix Figure A6 (p.18) allows for specific slopes in income by types of “household composition” (i.e., number of adults X number of children under 18 X marital status). This figure establishes that, even if these households have different marginal utility functions $u'(c, \theta)$, we would still find self-targeting on MU within these household types on average.
 - f) Appendix Figure A10 (p.52) estimates self-targeting by health status (poor or fair, good, very good or excellent) by and labor hours (nonemployed, part time, full time). We find self-targeting across levels of health status and levels of labor hours. As a result, we don’t need to take a stand on whether such state dependence / nonseparability exists; our claims of self-targeting on “need” (marginal utility) would go through anyway.
 - g) In our reply to Comment 7, we walk through other tests that strongly suggest self-targeting on need even if one is skeptical of our consumption measure (e.g., we find self-targeting on low education). These tests are also responsive to the concern raised in Comment 8, since they suggest self-targeting on need even if the consumption to marginal utility relationship is not monotonic.
- 3) Even if the Reviewer remains unpersuaded that consumption is a “sufficient statistic” for marginal utility, consumption is key input to / proxy for welfare in most economic models. And on magnitudes, our results make it difficult to believe that “need,” measured in any plausible way, would not predict take-up. We do not make this interpretive point explicitly in the paper, but we are happy to add it if the Reviewer adds.

9. “There are two important measurement error issues in the use of PSID pretax, pretransfer annual reported income as a measure of need. One is that the annual income measure in the PSID—even if reported accurately—is not what agencies use to establish eligibility and benefits. In the simplest cases, they use monthly income, which is different than PSID annual income, and it is very likely that households participate in programs only some months in a year. But it’s even more complicated than that, because most transfer programs use some combination of past monthly income, an estimate of future income, etc. to calculate eligibility. It is almost impossible to capture this with the PSID. Now, it may in truth be the case that program operators collect low quality income data when they assess eligibility. True monthly income may equal monthly consumption exactly, for example. In that case, the policy implication is that transfer programs should do a better job of collecting income data (e.g., using register data, etc.).”

We thank the Reviewer for this point. We appreciate their concern that transfer eligibility and benefits are not truly determined in a way that neatly aligns with the annual concept of most PSID variables.

We revised Section 3.4 (“Sensitivity to Mismeasurement”) to discuss this critique explicitly (p.21) and added two new robustness checks (Appendix Figures A8 and A9 on pp.50-51):

“Third, we are mostly limited to annual data, despite the possibility of correlated within-year fluctuations in receipt, eligibility, income, and consumption. For three transfers, however, we can

show that measuring receipt in terms of the share of months in the year, or comparing only zero-month to twelve-month households, matters little for our PSID results (Appendix Figure A25). In addition, monthly data on food insecurity and SNAP take-up allow us to test directly for aggregation bias from annual data (Appendix Table A5). Finally on time aggregation, we also find similar self-targeting when we look only at households without any change in employer in the last two years (Appendix Figure A26). Such households are likelier to have stable incomes, reducing one source of potential aggregation bias from annual data.”

The essence of our strategy here is, while we cannot make the perfect data “appear” in the PSID, we can look at subsamples where the data concerns they raise are likely to be absent or relatively limited. Here, the sensible populations to consider are the year-round continuously enrolled and the year-round continuously employed. Had we seen less self-targeting among the continuously employed (or continuously nonemployed) than among those with substantial within-year income volatility, for example, it would very much raise the Reviewer’s concern that annual aggregation was making within-year eligibility/income variation look like self-targeting. So we view our findings as quite meaningful.

Monthly data on food insecurity in the PSID allow us to run comparable “self-targeting” regressions at the monthly level and annual level. We find similar estimates from those regressions, and a regression with person-year fixed effects finds little self-targeting on within-year variation in food insecurity (Appendix Table A5, p.80).

It is worth noting that most transfer-months occur within long spells on transfers, despite substantial churn in these programs (see Bane and Ellwood 1986).⁶ This fact helps explain why self-targeting estimates are relatively insensitive to within-year churn (as we note on p.68, alongside Appendix Figure A25): “Partial-year recipient households respectively account for 4, 17, and 21 percent of receipt-months in SSI, SNAP, and TANF respectively.”

10. “The other obvious problem is that PSID respondents may underreport Y income to the PSID interviewers, either intentionally or unintentionally. There are many papers which match administrative earnings data to survey data on earnings and find large differences (e.g., Bollinger et al., JPE, 2019). This paper addresses that problem by using predicted Y in place of actual Y, using a prediction equation from a different data set (the CPS). So it uses IV, which is a textbook approach to correcting for measurement error. However, IV requires that all the variables used to predict are not direct determinants of the outcome in question (consumption) and that they are exogenous. The variables used as instruments here--hours of work, weeks worked, self-employment status, etc.)--are unlikely to satisfy either condition. A much more sophisticated treatment of measurement error in income is needed.”

We thank the Reviewer for raising the important issue of measurement error in income. We have three responses:

⁶ Bane, M.J. and Ellwood, D.T., 1986. Slipping into and out of Poverty: The Dynamics of Spells. The Journal of Human Resources, 21(1), pp.1-23.

- 1) The manuscript now includes a new sensitivity analysis of self-targeting with respect to measurement error in income (see p.23 in the paper, “our constrained regression” approach). Our memo reviews this analysis in reply to this Reviewer’s Comment 4.
- 2) We revised the paper’s discussion (p.23) of what kinds of income underreporting would/wouldn’t present a source of bias to our estimates of self-targeting. The first sentence is new:

“The underreporting of income on surveys would affect our analysis only insofar as transfer recipients and nonrecipients have different underreporting rates. In fact, one plausible story of income misreporting runs counter to our results. In this story, transfer recipients have incentives to underreport income to maintain eligibility. They may do so in any quasi-official setting, including surveys. These incentives might apply less strongly to consumption and to nonrecipients. All else equal, transfer recipients would thus appear positively selected on consumption given income.”

We are not familiar with direct evidence on heterogeneity in income reporting rates.

- 3) We wish to clarify that we do not use IV for the predicted-income strategy. We apologize for any confusion due to our writing. We completely agree that such an approach would not work. Instead, as noted in the original draft (p.18), we “control[] for predicted income, in addition to reported income” and find this “has a modest impact on our results.” That is, we augment our self-targeting regression with an additional control for predicted income, rather than using predicted income as an instrument for PSID income. The rationale for this control strategy is straightforward: using two noisy proxies for income provides better measurement than relying on a single proxy. We also acknowledge limitations of this strategy: “While this method cannot completely address measurement error, it mitigates the concern by effectively consolidating many labor variables into additional controls.”

We welcome any other suggestions from the Reviewer about how to better handle the threat posed by income mismeasurement, whether by improving our existing strategies or suggesting alternatives.

11. “Table 1 take-up rates are lower than in official and other estimates. The paper notes this and has some of the alternate estimates in Appendix Table A10. But no explanation is given for the difference.”

We thank the Reviewer for this opportunity to review our paper’s current discussion of this point:

- 1) We observe, on p.9 in the discussion of Table 1, that our estimates of “take-up rates are lower than official estimates.”
- 2) In a footnote on this page, we then summarize our view of the discrepancy. This footnote is unchanged from our original manuscript:

“Appendix B provides two supplementary analyses that probe why our estimated take-up rates are lower than figures published by government agencies. First, focusing on SNAP,

we decompose the gap into three components: differences in receipt, differences in simulated eligibility, and a “misalignment” component. This third component reflects a little-discussed flaw in official figures that biases them upward: As they simply divide total enrollment by total simulated eligibility, there is no adjustment for recipients in the administrative data who would be simulated ineligible in survey data. We find the underreporting of receipt (see Section 3.4) and the misalignment issue are both quantitatively important to the gap in take-up rates. Second, we review estimates of take-up rates for other programs. Our estimates appear in line with other survey-based estimates but are below administrative-data estimates, consistent with transfer underreporting and misalignment.”

- 3) Appendix B.6 (pp.102-103) provides an extended discussion of the reasons we are finding a lower take-up rate, including all mentions of Appendix Table A16 (previously A10). We quote it below in full, noting that there have been some revisions to it in response to comments from Reviewer 3:

In this section, we examine why take-up rates in Table 1 are smaller than official estimates. Here, we define take-up rates to be the share of households that take up, among households we simulate to be eligible. Receipt rates are the share of households that take up, among all households.

Summary Statistics. Appendix Table A14 reports sample-average estimates of take-up rates as well as three other quantities: (1) receipt rates among the simulated-ineligible, (2) rates of simulated eligibility among recipients, and (3) rates of simulated eligibility among nonrecipients.

Take-Up Rates in SNAP. Appendix Table A15 compares our estimates of take-up rates to USDA official estimates. Ours are clearly lower than the USDA estimates.

One explanation for the discrepancy is that, without conditioning on simulated eligibility, a lower share of households report that they receive SNAP in the PSID than are known to receive SNAP in administrative data. The implied rate of underreporting of SNAP benefits is roughly consistent with prior research (e.g., Meyer et al., 2009). Underreporting of receipt can thus explain some of the difference in take-up rates. We see this as encouraging: A substantial literature has developed around this form of measurement error in take-up rates, and we use corrections for underreporting developed in Mittag (2019) as a robustness check (see Section 3).

A second explanation for low take-up rates in the PSID relates to how eligibility enters official estimates. Appendix Table A15 shows two methods of computing SNAP take-up rates: (1) the ratio of total receipt to total simulated eligibility and (2) the receipt rate among the simulated eligible. The USDA takes the first approach, which is accurate only if every recipient is simulated ineligible. We avoid that assumption by taking the second approach in our main analyses.

By applying both approaches in the PSID, we can offer novel evidence on this potential contributor to the discrepancy in estimates of take-up rates. Appendix Table A15 shows that, when we follow the first approach, we move toward the USDA take-up rates. The second approach, however, yields a lower take-up rate due to what we call “misalignment”: Many recipient households in administrative data are likely to be simulated as ineligible if they were to appear in the survey data. Such households would enter the numerator but not the denominator of the USDA estimate, and this appears to be a key factor in the discrepancy. For this reason, official estimates of take-up rates ---which divide the administrative data in the numerator by simulated eligibility estimates, without adjusting for false negatives in eligibility---likely overstate true rates.

The most striking example of this upward bias is when the first approach yields estimates exceeding 100-percent take-up. For instance, in fiscal year 2018, the USDA reported take-up statistics with such issues in four states: DE, IL, OR, WA.⁷ Furthermore, in all of these states in that year, there were specific demographic cells with official reported take-up rates of less than 100 percent, which is mathematically inconsistent with 100-percent take-up overall. In our view, these challenges suggest that survey-data measures of take-up are not obviously dominated by approaches leveraging administrative data.

For further external evidence on this “misalignment” problem, see Newman and Scherpf (2013) and McBride et al. (2023). Using data linked from surveys to administrative records, they observe receipt and simulate eligibility for the same people, studying SNAP and WIC respectively. Both find take-up rates well below official USDA estimates.

Benchmarking Other Programs. Appendix Table A16 reports a selection of estimated take-up rates in other research. These estimates come from a variety of data sources (administrative and survey) and time periods. Overall, our estimates of take-up rates are lower than administrative estimates, but they appear generally consistent with other survey-based estimates of self-reported receipt among simulated eligibles. For these programs, the most plausible explanation for low take-up rates is survey underreporting of transfer receipt, as in, e.g., Meyer et al. (2009).

- 4) We also wish to make note of the several checks the paper already contains regarding underreporting of transfer receipt as a potential source of bias in estimates of self-targeting; see p.22.

12. “Housing is a special case: it is not an entitlement program because only a certain number of housing units and vouchers are funded by the government. It is rationed, and there are long waiting lists for a unit or voucher. Yes, self-selection can occur by who applies and who does not, but the government has its own priority list for who to select from the queue. If it is the government that is

⁷ See <https://www.fns.usda.gov/usamap/2018>. [Footnote appears in the paper.]

doing positive selection, perhaps the policy recommendation is that all programs should be rationed and we should let the government decide who is allowed to participate.”

We thank the Reviewer for raising housing assistance as an interesting case to consider. We engage with this point only in the R&R memo, but we are ready to include some of the following discussion in the paper or appendix if useful.

- We view priority lists as some combination of eligibility rule and ordeal. For example, the household’s (observable-to-government) characteristics determine its waitlist priority level. At some priority levels, the household will face some wait time (which is analogous to our ordeals) or might be outright excluded (so essentially ineligible). By implication, we see the rationing of housing assistance as well encompassed within our framework. If the Reviewer wants, we can add a footnote acknowledging that, in housing, the distinction between eligibility rules and ordeals can be unclear.
- In our view, the most persuasive counterargument to the reasoning just advanced is: Housing assistance is effectively a different program at each priority tier, due to the difference in wait times. Fortunately, in Panel A of Appendix Figure A27 (p.70), we continue to find self-targeting in housing assistance after we control for many household characteristics other than income-rank. This suggests that, if we could compare households that face different *potential wait times* if they were to apply for housing assistance (an infeasible regression, since we do not have data on the queues), we would still find self-targeting on average from these within-wait-time comparisons.
- Waldinger (AER 2021) presents evidence of precisely this self-targeting claim (transfer is valued more highly by needier people, conditional on wait time) by estimating a structural model of household preferences for public housing in Cambridge, MA.⁸ He is able to control for income in that analysis and finds remaining selection on other predictors of need, such as race/ethnicity.

⁸ Waldinger, D., 2021. Targeting in-kind transfers through market design: A revealed preference analysis of public housing allocation. *American Economic Review*, 111(8), pp.2660-2696.

Reviewer 3

We thank the Reviewer for their constructive suggestions. We wish to make one preliminary note before responding in detail to the Reviewer's comments. Many of their comments relate to the introduction and discussion of our self-targeting regression framework. We have completely rewritten Section 3.1 (pp.11-13) in light of their points. The Reviewer may find it easiest to read those pages in full, and then to consult the comment-by-comment notes.

1. **“Test for average self-targeting: The coefficient on transfer receipt, β is the central statistic of the paper, so what it captures, how it should be estimated and what could go wrong should be very clear. The discussion is brief, hard to follow and some key issues are either skipped or found in other places. Some things just need to be explained more clearly or in the right place, here are two broad issues I would find important to clarify: First, it should be clear what it measures (average difference in the dependent variable between recipients and eligible non-recipients with the same income and X) and why $\hat{\beta} - \beta$ is a good statistic to test for self selection. Some specific issues that I found unclear are as follows: if I understand correctly, $\hat{\beta} - \beta$ only seems to measure the benefits of self-selection, but not the costs in terms of hassle and administration. So it can only tell us whether it can be worth it, we need an assessment of the costs to decide. I also think $\hat{\beta} - \beta$ captures a combination of self-selection and unmeasured eligibility criteria since some of those predicted eligible may be rejected applicants. Another issue I would like to see commented on is what kind of welfare criterion your test implies – as far as I understand, by weighting people equally, it ignores hassle costs and does not care about whether self-selection is more prevalent among the most needy or the marginal applicant. I believe these are reasonable choices and the test could be modified to reflect other choices by using weights in the regression.”**

We thank the Reviewer for their comments here. We are unfortunately not confident we understand their reference to $\hat{\beta} - \beta$ versus $\hat{\beta}$, as we do not make a distinction of this kind in the paper. However, our best guess is that they are saying that our target parameter is “pure selection,” purged of any behavioral responses to the program, when they mention $\hat{\beta} - \beta$. Meanwhile, $\hat{\beta}$ represents the regression estimate of selection, which also is “contaminated” by behavioral responses. Our comments below proceed with this interpretation. We sincerely apologize for our confusion and invite the Reviewer to clarify if we are mistaken.

Under the above assumption, we made several changes in response to these suggestions:

- 1) We revised our statement of what the regression coefficient is on p.12: *“The coefficient β is therefore the average difference in consumption rank or lifetime-income rank between transfer recipients and eligible nonrecipients with similar incomes.”*
- 2) The Reviewer is correct that Section 3 measures the benefits, not the costs, of self-targeting. As they state, the benefits will need to be compared to costs (using modeling assumptions, as we do

in Section 4). We now say so explicitly at the start of Section 3 (p.11): *“In turn, these measurements of self-targeting yield our estimated social benefits of take-up, which we will compare to estimated social costs of ordeals.”*

- 3) The Reviewer then raises the issue of unmeasured eligibility criteria. In discussing the regression approach in Section 3.1, we added new text on this point (p.13):

“Fourth, the ideal estimate of self-targeting compares recipients with eligible nonrecipients, but we only have imperfect measures of simulated eligibility. By implication, our sample will inevitably exclude some true-eligibles and include some true-ineligibles. Measurement error in eligibility is, moreover, unlikely to be independent of consumption. The exclusion of some eligibles and the inclusion of some ineligibles are thus potential sources of bias.”

Our robustness checks regarding unmeasured eligibility criteria can be found on pp.20-22. In overview, we do four things there: [1] show how our estimates would change if we tweaked the eligibility criteria, for example by changing income limits or adding asset tests; [2] show robustness to reclassifying all simulated-ineligible recipients as simulated-eligible; [3] estimate self-targeting exclusively among demographic cells that are extremely likely to be eligible; [4] estimate self-targeting exclusively among households with stable employment and take-up status; [5] we use an Oster (2019) test to show that unmeasured eligibility criteria would have to be extremely predictive of consumption/lifetime income, in fact, much more so than household characteristics like race, education, or marital status.

- 4) The Reviewer then asks what sort of welfare criterion our self-targeting test implies. When we put marginal utility as the dependent variable, the unweighted regression implies a utilitarian criterion. The self-targeting regression can also accommodate weighted utilitarian frameworks. The choice of social welfare function determines the regression weights, or equivalently, the marginal-utility dependent variable. For example, if we chose a highly progressive social welfare function, this would weight low-consumption households highly.
2. **“A second important issue to discuss is your choice to use $\hat{\beta}$ rather than $\hat{\beta} - \beta$ as the test statistic. On p. 12 you transit from one to the other without even mentioning it. You later discuss behavioral responses, but that’s much later and it is still not clear to me whether you assume that β is zero or whether you assume that $\beta > 0$, so that $\hat{\beta} < 0$ still implies advantageous self-selection. While I find your arguments regarding changes in consumption to be eligible reasonable, I think $\beta = 0$ is a very strong assumption. Behavioral changes in order to meet eligibility criteria do tend to be small, but there could also be an effect of the transfer on consumption or a correlation between preferences for take-up and preferences for consumption that would lead to $\beta \neq 0$. This issue is amplified by the problem that one can never use a complete measure of consumption as the dependent variable, so that the assumption would also fail if program participation is associated with differences in consumption patterns or has an effect on consumption of measured vs. unmeasured**

consumption. That makes is a bit less clear whether β may not be negative, so it may not be as easy to infer advantageous self selection. Yet what is more important is that even if $\beta \geq 0$, the test becomes “one-sided” in that it can only detect positive self-selection, but cannot indicate a problem of negative self-selection. It also seems to make $\hat{\beta}$ of very limited use in the welfare analyses below.”

We thank the Reviewer for their comments. We made two key revisions to the text in response to this comment:

- 1) We created a new subsection, “Consumption as a Measure of Need” (p.18), which we quote here in full. This subsection presents two new analyses (Appendix Table A4, p.79; Appendix Figure A3, p.45) and repositions a third check in the text (Appendix Figure A4, p.46):

“We have so far described our results in ways that imply take-up is a ‘tag’ of some exogenous concept of need. Behavioral responses to taxes and transfers, however, suggest that household need is not fully exogenous. Three pieces of evidence suggest take-up does indeed reveal information about households’ exogenous resources.

*First, we find self-targeting in the most plausibly exogenous correlates of need (see **Appendix Table A4**). In most programs, take-up among the eligible is strongly predictive of high-school dropout, single parenthood, disability, minority race and ethnicity, and lower rates of savings. If these groups are high-need, then these patterns also provide support for the assumption that low consumption reveals marginal utility.*

*Second, we show that consumption patterns in the years around take-up are consistent with self-targeting on persistent earnings ability, rather than behavioral responses to transfers (see **Appendix Figure A3**). Among current eligible nonrecipients of a given transfer, future recipients have on average a lower current-consumption rank than similar future eligible nonrecipients.* We find strong self-targeting even when one looks at take-up in the very distant future (10--20 years), ruling out strategic reductions in consumption just before transfer take-up.*

*Third, we show that the behavioral responses in consumption required to fully explain our results are very large (see **Appendix Figure A4**). This responds to the concern that recipients adjust reported or actual consumption in response to transfers. For most transfers, the required response exceeds one dollar for every dollar of transfer received. In addition, our focus on the targeting of the ‘last dollar’ of transfers is conservative. Receiving transfers itself enables households to raise consumption and thus lowers imputed marginal utility, and so the ‘first-dollar’ targeting of transfers would thus be even better than in our main results.”*

* The regression of interest is $\bar{R}_{it} = \alpha_{ct} + \beta D_{i,t+k} + f(R_{it}) + u_{it}$ within the subsample such that $D_{it} = 0$ for $k > 0$. [Footnote appears in the paper.]

- 2) We also revised Section 3.1 to cut the discussion of causal effects that was on p.12 of the original draft. We think this discussion was misleading and, in particular, suggested that our test required us to disentangle behavioral responses to transfers from selection. To be entirely clear, this is not required, and we apologize sincerely for any confusion that our writing created.

As in Chetty (2006), in our model, differences in marginal utilities between recipients and nonrecipients are sufficient statistics for the benefits. Behavioral responses affect the social costs of transfers, not their social benefits. They matter exclusively through their fiscal externalities. We already capture some fiscal externalities in Section 4 (fiscal effects of higher take-up and less labor supply). For this reason, any adjustment to the measure of social benefits would be, in essence, “double-counting” the costs and thus conceptually incorrect.

3. **“There are also a few issues that are central to implementing and interpreting the test, that I think the paper should address. The first issue is the choice of dependent variable and its consequences. You construct it very carefully, but it would be useful to understand what the ideal measure would be (individual welfare/utility?), how feasible measures differ from that and what that implies as well as why your consumption-based measure is ideal among feasible choices. These questions are likely very similar to common questions of resource measures for poverty (raising questions such as how to deal with savings, investments, etc.) and measuring well-being by consumption (raising questions such as how measures are affected by shifts in consumption and being unable to measure all consumption). There is no need to reinvent that wheel, but since choosing the dependent variable is a key part of the test, readers should be aware of the relevant questions.”**

We thank the Reviewer for these comments and, as mentioned, thoroughly revised Section 3.1 to address them. Here are the specific relevant changes:

- 1) We begin the section exactly as the Reviewer recommends, with a definition of the ideal quantity to be estimated. See p.11:

“The ideal measure of self-targeting in a given transfer is the average difference in the social marginal utilities of income (“need”) between the transfer’s recipients and eligible nonrecipients who are otherwise identical to the government:

$$[E[u_{c,1}] - E[u_{c,0}]] / E[u_c],$$

where $u_{c,1}$ and $u_{c,0}$ respectively indicate the marginal utilities of recipients and eligible nonrecipients. The denominator, average marginal utility, expresses this difference in money-metric terms. That is, the measure in Equation 3 captures a society’s estimated willingness to pay (per dollar) to redistribute resources from eligible non-recipients to recipients, under the assumption that fiscal externalities from this action are offset via lump-sum taxes on all eligible individuals.”

- 2) As the Reviewer recommends, we reference the literature on poverty measurement in introducing how we intend to go from the ideal comparison of marginal utilities to the regressions we actually run (pp.12):

“Translating the concept of marginal utility differences in Equation 3 into an empirical framework like Equation 4 involves several key choices. Many of these choices have parallels in the literature on measuring poverty and material hardship (e.g., Citro and Michael, 1995). Here we discuss four worthy of focus: the choice of consumption as a proxy for need, the use of rank-based measures, the control for current income, and the focus on simulated eligibility. Section 3.4 further explores the robustness of our results to these choices.”

- 3) As the above quote suggests, the first choice we discuss follows the Reviewer’s suggestion to argue “why your consumption-based measure is ideal among feasible choices” (see p.12):

“First, marginal utilities of income are unobserved, requiring a model to link the data to marginal utility. Here, we take consumption as a proxy for marginal utility, as others have commonly assumed (e.g., Finkelstein et al., 2019; Deshpande and Lockwood, 2022). Two broad motivations for this choice are the partial insurance of household consumption against income shocks and the mismeasurement of low incomes in both survey and administrative data. This choice aligns with consumption–savings models where marginal utility is monotonic in consumption. Under this monotonicity, consumption ranks align with marginal-utility ranks in a cross-section of households. Lifetime-income ranks would also align in this sense if consumption is fully insured against income shocks.”

- 4) We also now present a series of robustness checks on p.18 that are motivated by theoretical objections to using consumption as a measure of marginal utility or need. In brief, these seek to demonstrate that self-targeting would hold even if, for example, the mapping from consumption to utility evolved with age. We review these additions in reply to the Reviewer’s Comment 8.

- 4. “The second issue is the choice of independent variables. It would be good to briefly justify why you only control for income. I think that is sensible, but one could reasonably wonder whether it wouldn’t be better to control for household structure rather than using equivalence scales, whether one should control for price levels (which will make the map from income to consumption vary across areas and could thus bias β).”**

We continue to quote from our full rewrite of Section 3.1, motivated by these comments. We also made edits to Section 2.1 (if we viewed their comment as about a data variable’s definition) and to Section 3.4 (to report the robustness checks implied by the comments). We proceed in order of the points raised here:

- 1) On controlling only for equivalized income rank in the baseline specification (p.13):

“Third, the inclusion of flexible controls for current income reflects the theoretical perspective that transfers are designed to complement, not duplicate, income taxation. Of course, real-world income tax schedules depend on variables other than income, most notably aspects of household composition (e.g., household size and marital status). Further controls can be warranted to the extent the government would adjust eligibility rules in transfer reforms.”

The Reviewer also suggests robustness checks in which we control for household structure. We now do so in a new robustness check, Appendix Figure A7 (p.49). There we define household structure as a “cell” defined by the number of adults X number of children X marital status (similar to the tax system or an equivalence scale), and allow for separate flexible splines in income rank for each household-structure cell. As we now report in the main text of the paper (on pp.18):

“[W]e find similar self-targeting with and without equivalence-scale adjustments (Appendix Figure A3) and when we control flexibly for household composition (Appendix Figure A17). By implication, our results are insensitive to the choice of equivalence scale.”

- 2) The Reviewer then asks about local variation in price levels. Following Moretti (2013), we collected state-year price index data from the ACCRA Cost of Living Index (for 1997-2008), supplemented by the U.S. Bureau of Economic Analysis Regional Price Parity estimates (for 2009-2019).⁹ We use these indices to deflate consumption and income as the Reviewer advises.

A new Appendix Figure A7 (p.49) reports the results. We find they are almost indistinguishable from our main specification. We mention briefly these results on p.18. However, we do not discuss the theoretical point of whether it is appropriate to make such an adjustment for local price levels.

We think it’s best not to discuss the issue conceptually in the paper for two reasons. First, it doesn’t matter empirically for our purposes. Second, it is theoretically very subtle and an area of active inquiry. In particular, local price index data do not themselves account for differences in natural amenities or nontraded goods, and with perfect spatial mobility, the local price level is the same everywhere. However, our paper is interested in marginal utilities, not average utilities being equated under a spatial-equilibrium condition. Marginal utilities need not themselves be equal across space, and spatial variation has to do with whether income is a substitute or complement to local amenities (see Glaeser 1998; Davis and Gregory wp).¹⁰

⁹ Moretti, E., 2013. Real wage inequality. *American Economic Journal: Applied Economics*, 5(1), pp.65-103.

¹⁰ Glaeser, E.L., 1998. Should transfer payments be indexed to local price levels?. *Regional Science and Urban Economics*, 28(1), pp.1-20. See Davis and Gregory (wp), “Place-Based Redistribution in Location Choice Models,” for a state-of-the-art discussion of why spatial variation in marginal utilities is very challenging to identify in spatial-equilibrium models.

5. Yet the more important point is that I do not understand how you compare people with the same income when recipients have additional program income by definition. Does income include these transfers? If not, then there is a clear reason why recipients should have higher consumption than non-recipients (what else are they supposed to do with the transfer). Yet if you include it, we may be worried that the marginal propensity to consume out of a transfer differs from the marginal propensity to consume out of income, which would bias $\hat{\beta}$. That's clearly true for in-kind transfers such as housing, but could also be a problem especially if programs indeed provide insurance value (i.e. we'd expect those who have already played that card to save more). For in-kind transfers, errors in their valuation would also bias $\hat{\beta}$."

We thank the Reviewer for this question and have two comments:

- 1) The Reviewer then asks "*how you compare people with the same income when recipients have additional program income by definition. Does income include these transfers?*"

On p.6, we write: "*We define household income as total annual income of the head and spouse before taxes and transfers, excluding other household members. Income includes labor, business, and capital income.*"

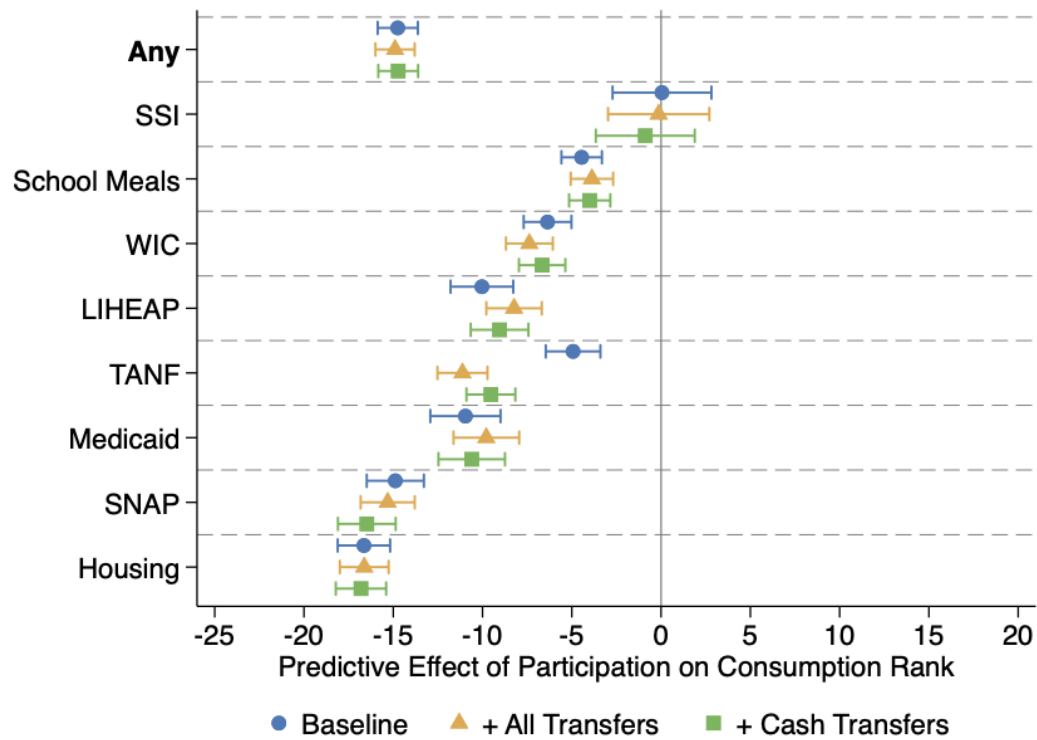
- 2) The Reviewer then notes that if income does not include transfers, then "*there is a clear reason why recipients should have higher consumption than non-recipients (what else are they supposed to do with the transfer). Yet if you include it, we may be worried that the marginal propensity to consume out of a transfer differs from the marginal propensity to consume out of income.*"

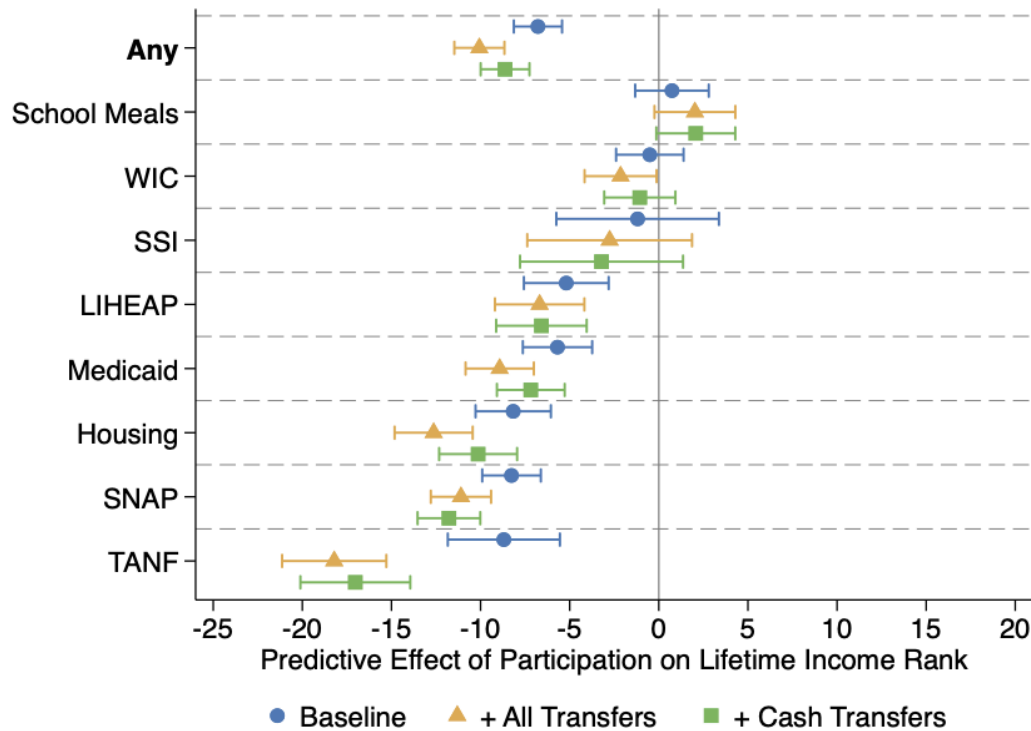
Our concept of transfer targeting is to compare households that are "ex-ante" equivalent to the tax system (i.e., have the same pre-tax pre-transfer income, which informs our choice of control), and then to compute the social value of providing an *additional dollar of transfers on top of what is already provided* (informing our choice of outcome variable). Other targeting concepts may yield other inclusion or exclusion rules from the consumption outcome or income control, but we are measuring what we seek to measure. Notably, none of the transfers in our study count toward taxable income, informing our choice to not count them as income.

An alternative targeting concept would be for the *first* transfer dollar. Importantly, this requires additional assumptions to estimate, most notably the MPC out of transfer income as the Reviewer mentions. However, we recognize one might want to have a rough estimate of "first-dollar" targeting. We now present such estimates Appendix Figure (A4, see p.46) which assumes an MPC of one, which is a salient upper bound. We summarize those results on p.17, writing:

"In addition, our focus on the targeting of the "last dollar" of transfers is conservative. Receiving transfers itself enables households to raise consumption and thus lowers imputed marginal utility, and so the "first-dollar" targeting of transfers would thus be even better than in our main results."

However, one might also wish to consider targeting concepts under the assumption that all transfer benefits, or cash transfer benefits, count as taxable income. We have prepared this analysis for the Reviewer and could include it as an appendix figure if desired. Please see below. Blue dots present our baseline estimates, yellow diamonds add the imputed value of all transfers to our measure of taxable income, green squares add only the cash transfers (SNAP, TANF, SSI, UI, and LIHEAP) to taxable income.





6. “I found it hard to follow some of the arguments in this part, maybe I am confused and it just needs to be explained (to me or in the paper). β seems to be something different in this part and its definition on p.25 does not seem to match the regression coefficient you use. You say β is the difference between the marginal social values of income between recipients and non-recipients. Yet above, β is the average difference between consumption between the groups. I do not understand how a regression of consumption on program receipt is informative about the value of income, much less the marginal value of income. Unless you make very restrictive assumptions, how can the same regression give you the average utility in the test above and the marginal utility here?”

We thank the Reviewer for raising this point, which led us to revise the manuscript in a few respects. The main thing we tried to clarify is that there is a “descriptive β ” and a “welfare-relevant β .” The “descriptive β ” comes from using consumption rank as the dependent variable, and the “welfare-relevant β ” comes from using the calibrated marginal utility as the dependent variable. Of course, constructing the calibrated marginal utility requires assuming a functional form for utility.

We want to be very clear that we are *not* taking the exact descriptive coefficient (from using consumption rank as the left-hand-side variable) and using that in the welfare analysis. The correct regression coefficient puts marginal utility on the left-hand-side, which is unobserved but, we argue, a monotonic transformation of consumption.

Previously, Section 3 showed only a descriptive β , and then our welfare analysis moved to marginal utility without an intermediate step. We now smooth the transition in several ways. First, we explain at p.12:

“To operationalize this concept of need, we begin with descriptive rank–rank regressions as a first step, followed by theory-consistent regressions. When we turn from descriptive to normative analysis, we replace the rank outcome $\bar{R}_{it}$ with calibrated values of marginal utility.” (p.12)

Then, on pp.15-17, we present a new analysis in which we calibrate marginal utility from consumption also gets us advantageous self-targeting on need. This regression leads to the estimates of welfare benefits we will use in Section 4. The similarity of the descriptive β and welfare-relevant β is unsurprising, since we assume marginal utility is monotonically decreasing in consumption, but we still see it as a useful transition into Section 4:

“Panel D connects our descriptive self-targeting regressions to the welfare analysis, and specifically to the welfare benefits of self-targeting. We do so by replacing the household-rank dependent variable with a calibrated value for each household’s marginal utility of income. The resulting coefficients can be interpreted as the additional value of moving a marginal dollar from a transfer’s eligible nonrecipients to its recipients.” (pp.15-17)

Above all, we hope that by including the theory-consistent regression estimates in Section 3 (as Panel D of Figure 1), we make it easier to follow the transition from rank-rank descriptive regressions to the welfare-relevant comparison. Thanks again for raising this point! We sincerely hope the manuscript now moves transparently from estimation to welfare.

- 7. I also struggled with some parts of the model setup and derivation. For example, your discussion of the optimal participation rate M on p. 30 suggests that you treat M as a choice parameter that can be chosen for a fixed elasticity and β . However, shouldn’t both the elasticity of participation and the average self-selection change with the participation rate? If so, there is a derivative wrt. to these parameters that enters the optimization problem, so one cannot optimize M holding the other two terms fixed. So maybe I am misunderstanding key parts of the model.**

We thank the Reviewer for flagging this discussion as needing revision. We have replaced the text with the following, which (in our view) avoids the implication that the elasticity and self-targeting coefficient are necessarily fixed with respect to the take-up rate:

“First, the difference in welfare between the reforms (dW/ds) varies in an intriguing way with the take-up rate. To illustrate this, observe that universal take-up ($M=1$) has no targeting benefits and that take-up is costly. On the other hand, take-up can also be “too low”: Inducing the first person to take up has no social value, as by the envelope theorem, their take-up cost exactly equals the benefits they receive. That is, only after the first person takes up are there any inframarginal households and thus is any first-order welfare gain possible. This suggests a

roughly U-shaped relationship between the welfare impact dW/ds and take-up $M(s)$, up to the labor-supply effect, as well as how β and ε_b might vary across levels of s .” (p.31)

Thus, we have cut the idea of the “optimal” take-up rate and are aiming more squarely at the intuition that there are downsides of both very high take-up (a loss of targeting from take-up) and very low take-up (few inframarginal recipients to serve). We invite the Reviewer to engage further, should this edit not address their concern.

8. **“I find the marginal analysis less compelling than the analysis of average self-selection for two additional reasons: First, in line with my struggles with β , the problem of analyzing consumption rather than utility seems more severe here. Utility is non-linear in consumption, so the marginal criterion will have an individual-level multiplicative factor that depends on the marginal utility of consumption of the recipients of the automatic vs. self-selected program. There is obviously little one can do about that, since we cannot measure utility. But mirroring my comment about welfare criteria in the average measure, I would expect some discussion of the consequences of replacing welfare weights with your consumption measure.”**

We thank the Reviewer for this thoughtful comment. It led us to make several revisions to the paper, which we now explain:

- 1) A focus of our revision was to show more of the steps in transitioning from measuring self-targeting on consumption to self-selection on marginal utility. A key revision was to put our model-implied measure of marginal utility directly on the left-hand side of our self-targeting regression (on pp.16-17, and see Panel D of Figure 1). We quote this in full:

“Fact 4: Marginal utilities of income appear higher among transfer recipients than among eligible nonrecipients with similar incomes.

Panel D connects our descriptive self-targeting regressions to the welfare analysis, and specifically to the welfare benefits of self-targeting. We do so by replacing the household-rank dependent variable with a calibrated value for each household's marginal utility of income. The resulting coefficients can be interpreted as the additional value of moving a marginal dollar from a transfer's eligible nonrecipients to its recipients. To calibrate marginal utilities for each household, we use the following preferences over consumption and labor hours from Greenwood et al. (1988):

$$u(c, \ell; \psi_i) = \frac{1}{1-\gamma} \left(c - \frac{\psi_i}{1+1/\eta} \ell^{1+1/\eta} \right)^{1-\gamma},$$

where c is the household's real annual equivalized consumption and ℓ is the annual labor hours of the household head or spouse. We will use the same preferences in our welfare analysis (Section 4) and discuss them further there. We express our results as a money-metric by rescaling each marginal utility using the population-average level.*

This carefully constructed regression yields coefficients which are exactly the object in Equation 3, that is, the left-hand side of the Baily–Chetty equation for the optimal level of social insurance (Baily, 1978; Chetty, 2006). We thus capture the social benefits of self-targeting, which Section 4 compares to calibrated estimates of the social costs of take-up. Panel D shows the estimates with and without a sample limitation to the simulated-eligible.

Panel D shows that, in some but not all transfers, the marginal dollar of transfer to recipients is worth substantially more than the marginal dollar to eligible nonrecipients. For instance, society would be willing to pay \$1.96 in resources to make a nondistortionary transfer of \$1 from the average SNAP-eligible nonrecipient to the average SNAP recipient. We estimate this Baily–Chetty premium to be 1.32 on average over the eight transfers. It arises because, as shown in the other three panels, transfers tend to attract take-up by households with higher “need” than is indicated by their income and eligibility status alone. Indeed, in transfers where we found no advantageous self-targeting on consumption and lifetime income, we do not find such a premium.”

** We calibrate $\gamma = 3$ (capturing both individual risk aversion and social welfare weights) and $\eta = 0.3$ as constants across households. The disutility parameter is calibrated internally as $\psi_{it} = w_{it} / \ell_{it}^{1/\eta}$, which we obtain by inverting the household’s labor supply function $\ell_{it}^* = \ell_{it}^*(w_{it}, \psi_{it})$. Following Finkelstein et al. (2019), we also impose a “consumption floor” at the fifth percentile in calculating marginal utility.
[Appears as footnote in paper.]*

- 2) Another point we see in the Reviewer’s comment is some concern that marginal utility could be a complicated transformation of consumption.
- Appendix Table A3 (p.76) shows that the empirical fact of self-targeting is highly robust to various functional form choices. There we use a Poisson regression to get the differences in percentages and in dollar levels, rather than in ranks, for consumption and lifetime income between recipients and eligible nonrecipients.
 - Section 3.1 raises the possibility of variation in preferences such that the mapping from consumption to marginal utility is not simple and monotone. On p.13, we now write that:

“Nonetheless, imputing marginal utility from consumption has inherent limitations. Perhaps the most significant is that it is infeasible to fully capture all variation in preferences when performing this imputation. For instance, factors like aging, parenthood (Blundell and Preston, 1998), opportunity costs of time (Aguiar and Hurst, 2005), and habit formation (Chetty and Szeidl, 2016) all influence the relationship between consumption and marginal utility.

Heterogeneity in labor-supply elasticities or nonseparability of labor and consumption create similar difficulties.”

- We then show robustness to what we saw as the most natural forms of taste heterogeneity. Our approach is to show that, within categories of households that are plausibly distinct in their tastes, we still see ample self-targeting into transfers. These results are all reported on pp.23-24 in the main text.
 - a. Appendix Figure A9 (p.50) estimates self-targeting by age group, and Appendix Figure A8 (p.51) allows the control for income to have a cohort-specific slope (so that the self-targeting coefficient can be interpreted as an average of within-cohort comparisons). These figures adopt the recommendation of Blundell and Preston (1998) for within-cohort comparisons of consumption when assessing welfare.
 - b. Appendix Figure A6 (p.48) allows for specific slopes in income by types of “household composition” (i.e., number of adults X number of children under 18 X marital status). This figure establishes that, even if these households have different marginal utility functions $u'(c, \theta)$, we would still find self-targeting on MU within these household types on average.
 - c. Appendix Figure A10 (p. 52) estimates self-targeting by health status (poor or fair, good, very good or excellent) by and labor hours (nonemployed, part time, full time). We find self-targeting across levels of health status and levels of labor hours. As a result, we don’t need to take a stand on whether such state dependence / nonseparability exists; our claims of self-targeting on “need” (marginal utility) would go through anyway.
9. **“Second, I am not sure the analysis captures the most relevant policy consideration. It seems relevant for the question whether one or another transfer program should be expanded, which is indeed relevant. Yet the more frequent question is whether we should increase or decrease hassle costs or eligibility criteria of a given program. As you mention, that is a very difficult question, since we do not know who get screened out by specific hassle costs such as stigma.”**

We thank the Reviewer for encouraging us to clarify our policy question.

Our hypothetical reform focuses attention on a key design choice: To what extent should welfare states rely on voluntary transfers versus automatic transfers? Of course, there are many ways one can transform welfare states to move toward or away from automatic programs. We focus on reforms that change transfer benefits, rather than hassle costs or eligibility.

Contemporary policy debates involve a mix of questions related to whether transfers should be automatic, their benefit levels, their hassle costs, and their eligibility criteria.

- In a footnote on p.1, for example, we draw on two pandemic changes between automatic and voluntary transfers:

“Before the end of Medicaid auto-enrollment, U.S. Department of Health and Human Services (2022) forecasted that 8.2 million people would lose benefits, of which 6.8 million (83 percent) were expected to be from non-take-up among the eligible. Some policy reforms have already occurred in the context of school meals: Ten states have extended the pandemic-era expansion as of Spring 2024 (see <https://frac.org/healthy-school-meals-for-all>).”

- More broadly, public finance economists have long been focused on changes in benefit generosity as a way to study the social safety net, going back to the Baily-Chetty formula (1978, 2006). A large body of work has leveraged this formula to estimate optimal benefit levels of UI, DI, and other programs. And there has been a lot of policy attention to benefit levels lately, including pandemic-era changes to the Thrifty Food Plan and UI benefit supplement.
- We also think our paper provides a useful contrast to studies of ordeal changes for several reasons that we review in the paper:

“Both ordeal changes and benefit shifts can offer valuable insights. Governments routinely consider changes in benefit levels as well as changes to eligibility and ordeals. Considering a larger space of transfer reforms has two other potential benefits. First, the literature finds mixed results on targeting under the first approach (e.g., Deshpande and Li (2019) versus Finkelstein and Notowidigdo (2019)), seemingly due to heterogeneity in marginal households across settings and ordeals. The resultant uncertainty gives reason to examine other policy hypotheticals, like our “voluntary-to-automatic” reform, which might (and indeed does) offer a more-robust empirical conclusion. Second, our analysis requires only observational data, since it is fundamentally descriptive, whereas identifying marginal recipients requires exogenous variation in ordeals.” (p.14)

“Third, our reform differs fundamentally from changes to ordeals. We instead take the ordeal as given and reallocate resources between voluntary and automatic transfers. Our welfare formula therefore weighs the value of transfers to inframarginal recipients against ordeal costs to marginal recipients that disenroll when the voluntary transfer is cut. By contrast, the welfare analysis of ordeal reforms weighs the change in ordeal costs to inframarginal recipients against the fiscal externalities from changes in take-up (Finkelstein and Notowidigdo, 2019). A virtue of our reform is that the welfare-relevant measure of ordeal costs is obtained by the envelope theorem. These are otherwise difficult to measure.” (pp.30-31)

10. **“Yet even if the question is about relocating money, an unattractive feature of your criterion is that the extensive margin of participation drops out in the marginal analysis. Unless I misunderstand, the criterion thereby captures the desirability of a reform where no one**

drops out of the self-selected program (since those who drop out are exactly indifferent, so they have point mass zero for continuous costs and benefits). By holding participation fixed, your criterion thereby implicitly seems to abstract from the very question you raise, which is the value of (changes in) self-targeting and instead evaluates whether adding a dollar to a given self-targeting regime offsets its costs (in terms of labor supply and hassle costs). In practice, the more relevant question seems to be how self-targeting changes with changes in the participation rate induced by changes in benefit amounts. For changes in benefit amounts that are not infinitesimal, whether those who drop out of the program are more or less self-selected clearly matters. The order in which people exit the program, which is difficult to infer, also affects whether the social welfare function is monotonic in the benefit amount. That we can neither assume monotonicity nor infer how β changes with M seems to make the sufficient statistic approach less attractive, since it can point away from the global optimum in these situations.”

We thank the Reviewer for raising these points. In what follows, we provide two broad answers. First, we describe the edits made to the paper that make clear we are focusing on a marginal reform. Second, we reply to the Reviewer’s comments about our marginal reform.

1) Edits to clarify focus on marginal reform and their limitations

- We revised the paper to precisely state our welfare question in the introduction (on p.3): “*When society allocates an additional dollar, should it be used to raise the benefit size of a voluntary transfer or to make an automatic payment to all eligible individuals?*” We agree that, for non-marginal changes in transfers, a variety of additional terms enter, including the selection properties of marginal recipients. However, we are not asking that question.
- We have edited the paper to acknowledge the limitations of marginal reforms, and the additional challenges of non-marginal ones, on p.31 and p.36:

“[W]e study marginal reforms, in which marginal recipients only matter for welfare through fiscal externalities, as they are indifferent to take up. However, in a non-marginal reform, people whose take-up choices change due to the reform are not necessarily indifferent. By consequence, who they are — that is, self-targeting among marginal recipients — would affect welfare.” (p.31)

“Our welfare test, while transparent, does not allow us to contemplate optimal transfer design or non-marginal reforms... These limitations of our study offer directions for future research.” (p.36)

2) Replies to specific comments from the Reviewer

- We appreciate the Reviewer’s perspective on the relevance of different types of reforms. Our approach aligns with a well-established literature analyzing marginal reforms, which remain a

central focus in public finance (e.g., Chetty 2006).¹¹

- The Reviewer suggests that the participation margin “drops out” or that we “hold[] participation fixed.” Even in the marginal analysis, the participation margin does not drop out, and participation changes. People who change take-up status identify the welfare costs, via the envelope theorem (see the first paragraph on p.32).
- The Reviewer then raises an important concern about “whether those who drop out of the program are more or less self-selected clearly matters.” Our envelope theorem argument shows the welfare-relevance of selection among marginal recipients is second-order: It will only have a quantitatively significant impact on a welfare analysis of a substantial change to transfer benefits; see Kleven (2021).¹² Modest changes to benefit sizes will then resemble the welfare analysis we perform. We added the above quotation (p.31) in part to acknowledge this as a possible limitation of marginal reforms.
- The Reviewer also suggests that our theoretical analysis “abstracts from the very question [we] raise.” In reply to this Reviewer’s Comment 6, the paper now explains how our welfare comparison is highly consistent with the empirical regressions we run. In Section 3, we estimate the selection properties of inframarginals, and these are what’s relevant for the marginal reform we consider in Section 4. By contrast, we are not in a position to evaluate non-marginal reforms. As the Reviewer suggests, we’d need to know the selection properties of marginal recipients. Since we don’t study any non-marginal transfer reforms, we cannot say who is marginal to take-up.

It is possible that we are misunderstanding aspects of the Reviewer’s concern, and we would be very grateful to have the chance to clarify. Thank you for pushing us to engage seriously with what we agree is a reasonable limitation of the paper.

11. “And even if they are indifferent in terms of balancing hassle costs and benefits of program participation – shouldn’t our valuation of a marginal change depend on whether the marginal individual is positively or negatively self-selected (in addition to the inframarginal dollar that your formula captures)? I guess I am somewhat puzzled by the fact that the derivative of β with respect to M drops out of the equation.”

We thank the Reviewer for the chance to clarify this point. Here we wish to provide a “toy” example of why, for benefit-size changes, marginal recipients only matter through their fiscal externalities and not through their selection properties (i.e., if they are needier or not).

- Suppose we increase the voluntary benefit size by a small $\delta > 0$ for all people, so that someone previously eligible for $S(z)$ may now receive $S(z) + \delta$. This amounts to an average transfer of $\delta M(z)$ from all people at income z , where $M(z)$ is the original take-up rate at income z . All of

¹¹ Chetty, R., 2006. A general formula for the optimal level of social insurance. *Journal of Public Economics*, 90(10-11), pp.1879-1901.

¹² Kleven, H.J., 2021. Sufficient statistics revisited. *Annual Review of Economics*, 13(1), pp.515-538.

these households are inframarginal to this reform, and assuming they value the transfer dollar-for-dollar, they see a utility gain of $\delta M(z)$. We can later weigh this using social welfare weights.

- In addition, there is also a behavioral response to this δ -increase in benefits. Given a take-up elasticity ε , take-up changes by $dM(z) = \delta \varepsilon M(z)/S(z)$ (obtained using the definition of an elasticity). Taking $\delta \rightarrow 0$, the marginal person who enrolls in response to the δ -increase is exactly indifferent between taking up and not taking up, so the change in their welfare is zero. We need only consider the fiscal externality they impose. This fiscal externality is

$$(S(z) + \delta)dM(z) = (S(z) + \delta)(\delta \varepsilon M(z)/S(z)) = \delta \varepsilon M(z) + O(\delta^2) \approx \delta \varepsilon M(z),$$

where the $O(\cdot)$ approximation captures higher-order terms that go to zero more quickly. That is, for a δ -increase in benefits, there is a fiscal externality of $\delta \varepsilon M(z)$ and no welfare gain to the marginals. It is important to see that this fiscal externality term should *not* be adjusted by any social welfare weights in terms of who takes up – these people don't see a utility change, this is a fiscal externality only.

- Now we can conduct a welfare analysis of the δ -increase in benefits. The gains to inframarginals are worth $\lambda \delta M(z)$, assuming a welfare weight of λ on the transfer recipients relative to the population as a whole. The costs, arising from fiscal externalities of marginals, are worth $\delta \varepsilon M(z)$.
- Thus, a δ -increase in benefits is desirable if and only if

$$\lambda \delta M(z) > \delta \varepsilon M(z)$$

which simplifies to $\lambda > \varepsilon$, noting that $\delta M(z)$ cancels out.

We are happy to add this toy example to the appendix if the Reviewer finds it helpful; we see it as encompassed within the existing example in Section 4.1.

12. **“The application is very carefully done and there is a lot of attention to detail, but it glosses over aspects of key relevance for its implementation. Some key choices do not seem to be explained. For example, you present a theory about income and consumption levels, but then jump to using income and consumption ranks in the application. At a minimum, that requires an explanation, but I think you need to use levels, since I do not see a useful welfare criterion that would allow you to use ranks.”**

We thank the Reviewer for encouraging us to explain the use of ranks. We have now included the following text in Section 3.1 (on p.13), where we explain such specification choices and new analysis we conducted to address the concern:

“Second, we focus on rank-transformed outcomes to provide a clear, interpretable summary of self-targeting. The common use of rank-transformed outcomes is motivated by their robustness to outliers and scale differences. In our context, one drawback is that rank-transformed outcomes implicitly weigh the outcome distributions differently than the outcomes in levels or marginal utilities. We later show similar results using both of those measures. By implication, little is lost in moving from descriptive facts, presented in ranks, to welfare implications that are not in rank form.”

For the Reviewer’s reference, the levels regression is Appendix Table A3 (p.76) and the marginal-utility levels regression is Panel D of Figure 1 (p.15).

13. **“Another key issue in implementation is the prediction of eligibility. As a simple practical matter, you should discuss how you deal with recipients whom you predict to be ineligible. Simply excluding them raises several conceptual problems (predictors of eligibility likely negatively predict consumption and your robustness check only partly accounts for that) and implies that you cannot say anything about self-targeting of the program, since you are only able to analyze a non-randomly selected subpopulation.**

But there also seem to be problems that are harder to address, since they stem from the fact that crucial information is not available or measured with substantial error. As you mention, predicting eligibility involves considerable error and it would be useful to discuss its extent and potential consequences/solutions. Without better data on eligibility, the extent of error in predicting eligibility (see Newman and Scherpf, 2013 for some evidence) seems to make implementation of the test very difficult. I do not understand your argument regarding the direction of this bias in footnote 15. It seems to be based on the idea of OVB, yet misclassification of eligibility induces a sample selection problem. So the bias depends on expected consumption in the 4 relevant groups (2 correctly classified and 2 misclassified groups).”

We thank the Reviewer for these points, which led us to refine our discussion of eligibility misclassification in several places in the paper:

- 1) We revised a sentence in Section 3.1, introducing our regression specification, to be clear about the sample definition. We now say (p.12): *“Our baseline specification limits the sample to the simulated-eligible ($E_{it} = 1$), thus dropping the simulated-ineligible.”* In the original manuscript, we said: *“The regression is estimated on the simulated-eligible ($E_{it} = 1$) unless otherwise noted.”*
- 2) In discussing the regression approach in Section 3.1, we added new text to discuss the sample-selection bias issue (pp.13-14):

“Fourth, the ideal estimate of self-targeting compares recipients with eligible nonrecipients, but we only have imperfect measures of simulated eligibility. By implication, our sample will inevitably exclude some true-eligibles and include some

true-ineligibles. Measurement error in eligibility is, moreover, unlikely to be independent of consumption. The exclusion of some eligibles and the inclusion of some ineligibles are thus potential sources of bias.”

- 3) We revised the problematic discussion (originally footnote 15) flagged by the Reviewer, which conflated a sample restriction and OVB. The relevant text (on p.20) now reads as follows:

“We mostly analyze self-targeting among people who are simulated-eligible for transfers, a group differs from the truly-eligible through both errors of inclusion and exclusion. The two-sidedness of the sample selection issue leads to a bias of ambiguous sign.”

- 4) And then, in the associated footnote, we write:

“Appendix C formally derives the bias of our estimator. We show the bias reflects a “difference-in-difference” term, comparing the predictive effects of transfer receipt on the outcome among the simulated-eligible versus the simulated-ineligible, along with a term containing the predictive effect of simulated-eligibility on the outcome mean. We argue the first term is of ambiguous sign but likely small magnitude, while the second term is likely negative. We conclude sample selection plausibly biases our estimator toward our finding of self-targeting. Many values in the bias formula are not well known, so we proceed with robustness checks rather than direct calibration of the formula.”

- 5) As the text suggests, we now provide a formal derivation of the bias of our estimator in Appendix C (pp.102-103). We will not quote the derivation here, for space reasons, but the result is much like the Reviewer suggests.

We hope these edits strengthen our treatment of the eligibility measurement issue. Other robustness checks in the paper include: (1) results with and without the simulated ineligible (p.11,p.16); (2) modifying income thresholds or asset tests has little effect on our estimates (p.21); (3) reclassifying the simulated-ineligible recipients as simulated eligible (p.21); (4) limiting the sample to very-likely-eligible people; (5) sensitivity checks for unmeasured eligibility rules in the spirit of Oster 2019.

We are highly motivated to make sure that we fully address this concern. We welcome any specific suggestions from the Reviewer if they think there is more we might do.

- 14. “Another issue is the measurement of income. As far as I understand, you use annual income both in the analyses and in the determination of eligibility. Yet program eligibility typically depends on monthly/current income. On the negative side, that leads to a lot of misclassification in eligibility. On the positive side, it means that your estimates of the insurance value omit the value of insurance against within-year variation in income, so that the insurance value of these transfers is even more impressive. In a similar vein, I am not sure how to think about your analysis given that participation is also monthly rather than annual. My impression is that one would need (much more accurate) measures of variables**

such as income, consumption, eligibility and program participation at the monthly level to implement the test.”

We thank the Reviewer for this point. We appreciate their concern that transfer eligibility and benefits are not truly determined in a way that neatly aligns with the annual concept of most PSID variables.

- We revised Section 3.4 (“Sensitivity to Mismeasurement”) to discuss this critique explicitly (p.21) and added two new robustness checks (Appendix Figures A8 and A9 on pp.50-51):

“Third, we are mostly limited to annual data, despite the possibility of correlated within-year fluctuations in receipt, eligibility, income, and consumption. For three transfers, however, we can show that measuring receipt in terms of the share of months in the year, or comparing only zero- month to twelve-month households, matters little for our PSID results (Appendix Figure A25). In addition, monthly data on food insecurity and SNAP take-up allow us to test directly for aggregation bias from annual data (Appendix Table A5). Finally on time aggregation, we also find similar self-targeting when we look only at households without any change in employer in the last two years (Appendix Figure A26). Such households are likelier to have stable incomes, reducing one source of potential aggregation bias from annual data.”

- The essence of our strategy here is, while we cannot make the perfect data “appear” in the PSID, we can look at subsamples where the data concerns they raise are likely to be absent or relatively limited. Here, the sensible populations to consider are the year-round continuously enrolled and the year-round continuously employed. Had we seen less self-targeting among the continuously employed than among those with substantial within-year income volatility, for example, it would very much raise the Reviewer’s concern that annual aggregation was making within-year eligibility/income variation look like self-targeting. So we view this test as quite meaningful.
- Monthly data on food insecurity in the PSID allow us to run comparable “self-targeting” regressions at the monthly level and annual level. We find similar estimates from those regressions, and a regression with person-year fixed effects finds little self-targeting on within-year variation in food insecurity (Appendix Table A5, p.80).
- It is worth noting that most transfer-months occur within long spells on transfers, despite substantial churn in these programs (see Bane and Ellwood 1986).¹³ This fact implies that self-targeting estimates are relatively insensitive to within-year churn (as we note on p.68, alongside Appendix Figure A25): “Partial-year recipient households respectively account for 4, 17, and 21 percent of receipt-months in SSI, SNAP, and TANF respectively.”

15. “And while I follow your arguments regarding survey error in program participation (though see minor point below), your arguments regarding measurement error in income do not address the most salient point: Even classical measurement error in income would lead

¹³ Bane, M.J. and Ellwood, D.T., 1986. Slipping into and out of Poverty: The Dynamics of Spells. The Journal of Human Resources, 21(1), pp.1-23.

to a negative bias in β (the coefficient on income is attenuated and income is negatively correlated with program participation, leading to negative bias), which could explain your results.”

We thank the Reviewer for pushing us to do more about measurement error in income. We have added a new sensitivity analysis to the paper that we find highly informative (on p.23), quoted here:

Next, we directly probe the sensitivity of self-targeting to any bias transmitted from income mismeasurement. In a linear version of Equation 4, we constrain the coefficient on income rank to higher values than the unconstrained estimate. We claim there exists a threshold value for the constrained coefficient that is implausible a priori, as it would imply an excessive noise share of income variance or unrealistically little smoothing of consumption relative to income. See Appendix C for further explanation of this procedure.*

Self-targeting in most transfers appears robust to substantial mismeasurement of income (see Appendix Tables A9, A10, and A11). In the PSID, when we constrain the rank–rank slope between consumption and income to 0.8, up from 0.51 when unconstrained, we find that SNAP recipients are on average 12.4 percentiles lower in the consumption distribution than eligible nonrecipients with the same income. This amounts to a downward adjustment of only 2.1 percentiles from our baseline estimate, despite the assumption of a substantial attenuation bias in the rank–rank slope.

[In footnote: $\bar{R}_i = \beta D_{it} + \gamma R_{it} + u_i$ where $\bar{R}_i$ is consumption rank or lifetime rank, R_{it} is income rank, and D_{it} is transfer receipt.]*

As the new text above suggests, we agree that, in principle, income mismeasurement is plausibly a source of bias. However, we show that self-targeting is robust to even substantial income mismeasurement. For example, in the text, we consider a case in which 40% of the variance in income ranks is noise (attenuating the coefficient on the income rank control from 0.8 to 0.51).

To provide a bit more econometric context on this sensitivity analysis: It recycles the idea of Anderson-Rubin IV confidence intervals. Anderson Rubin CIs are constructed by finding the interval of β at which the coefficient on γ is zero: $y_i - \beta x_i = \gamma z_i + e_i$. This should already look like our case: We’re constructing the interval of β (coefficient on income rank) consistent with $\gamma = 0$ (moment condition that transfer receipt has no direct effect). To see the connection with our work, suppose transfer receipt would not directly predict consumption rank given a perfectly-measured income rank. Then receipt is a valid instrument for the mismeasured income rank. Thus, if there is no sensible coefficient on income rank at which the coefficient on transfer receipt = 0, then transfer receipt cannot be an instrument. By implication, our finding of self-targeting couldn’t be explained by income mismeasurement.

Of course, we welcome specific suggestions from the Reviewer if they feel we have work left to do on income mismeasurement. Our other robustness check on this issue (Appendix Figure A28, p.71) is to supplement the income control with a control for “predicted income” constructed using hours, occupation, industry, etc. We find this does not greatly reduce our estimates of self-targeting.

16. **“Some of the points above clearly arise from data limitation and you do an impressive amount of work to address them. I would prefer the paper to discuss these remaining shortcomings and what kind of data/methods would be required to address them, because doing so would allow researchers to improve and better apply the very useful test you propose.”**

We thank the Reviewer for this suggestion and have made edits, largely to the introduction and the conclusion, to be more cautious and transparent about the data limitations. Many of these edits were also motivated by their Comment 20.

- Section 3.1 (pp.11-13), as previously noted, provides a high-level discussion of the challenges in implementing an empirical version of a theoretical test of self-targeting. Please see our replies to this Reviewer’s earlier comments.
- We fully rewrote the conclusion (p.35) so as to embrace the Reviewer’s “proof of concept” view (mentioned explicitly in Comment 20). Here is the new conclusion in full:

“This paper studies the positive and normative implications of “self-targeting” in U.S. transfer programs, in which eligible people self-select into take-up on the basis of need. We find substantial self-targeting on consumption and lifetime income across eight U.S. transfers. That is, transfer recipients appear needier on average than eligible nonrecipients, even when holding income fixed. This result is surprising in light of prior evidence, which has reached mixed conclusions as to how changes in take-up costs (“ordeals”) have affected program targeting. Combining our estimates with a new social-welfare formula, we estimate that in some transfers, the benefits of using take-up as a targeting device may plausibly exceed its costs.

Several limitations mean this study is best understood as a first effort to evaluate whether self-targeting can rationalize the widespread use of voluntary take-up in U.S. transfer programs. Among them are the difficulties of measuring income, consumption, transfer receipt, and transfer eligibility in our data. In addition, we have relied upon an adjusted measure of household consumption as an imperfect proxy for marginal utility. Our welfare test, while transparent, does not allow us to contemplate optimal transfer design or non-marginal reforms. Finally, we have abstracted from many dimensions of transfers. Among the ignored aspects are administrative costs, other ways to acquire information (e.g., disability reviews), and some aspects of household heterogeneity (e.g., selection on elasticities). These limitations of our study suggest directions for future research.”

- We replaced the final sentence of our abstract to be less decisive. It now reads: “*In some transfers, these social benefits of self-targeting appear to offset the social costs of take-up, challenging utilitarian arguments against ordeals.*” Before it had read: “*With a new sufficient-statistics result, we value the social benefits of self-targeting at approximately nine*

cents per transfer dollar, offsetting the social costs of take-up in some programs.” The caution occurs now in “*appear to offset*” and in avoiding a precise quantitative statement in the abstract.

- We cut several sentences from the introduction that were arguably overconfident, and we replaced them as follows.
 - Cut: “*We establish that take-up costs induce desirable and robust self-targeting on average,...*” Replaced with: “*Take-up appears to achieve desirable self-targeting on average...*”
 - Cut: “*Our results thus establish self-targeting as a credible argument for existing U.S. transfers and cast doubt on the merits of going automatic.*” Replaced with: “*Our results thus help advance self-targeting as an argument for existing U.S. transfers and against going automatic.*”

The referee suggests mentioning properties of the ideal dataset for our exercise. We are happy to add specific content discussion if that would be helpful.

17. **“Misreporting of program receipt is a key problem for your test. You clearly point this out and do a lot to address it. While I share your conclusion, I am not sure you make the most convincing case for the validity of your test in the presence of misclassification. After all, several papers have demonstrated misclassification rates high enough to overturn your results. However, the first order effect of misclassification should be to attenuate your test statistic, simply because any random misclassification will make recipients and non-recipients more similar. This tendency could be overturned if misclassification is correlated with consumption. My intuition is that it would require a peculiar setup for this to generate spurious results - those who do not report true receipt need to have higher consumption than those who truly report receipt (which is plausible) as well as than eligible non-participants (which seems less likely).”**

We thank the Reviewer for their praise for our detailed analysis of misreporting. In review, we provide three main pieces of evidence as to why we do not think transfer misreporting is a plausible explanation for our results:

- 1) The paper makes precisely the argument this Reviewer proposes (on p.22):

“[C]orrections consistently increase our estimates of advantageous self-targeting (Appendix Table A6)...[U]nder constant misreporting rates, our estimates are attenuated. Consider misreporting probabilities $p_0 = \Pr(\tilde{D}_i = 0 \mid D_i = 1)$ and $p_1 = \Pr(\tilde{D}_i = 1 \mid D_i = 0)$, where D_i indicates true receipt and $\tilde{D}_i$ indicates reported

receipt. Comparing the feasible regression of $y_i = \tilde{\beta} \tilde{D}_i + u_i$ to the infeasible regression $y_i = \beta D_i + u_i$, one can show that $\beta = \tilde{\beta} / (1 - p_0 - p_1)$.

We then leverage recent research on demographic predictors of transfer misreporting, which allows us to rigorously test the intuition they offer. We find that the Reviewer correctly observes one would need a “peculiar setup” in which misreporting is correlated with household consumption; that is, the recent corrections actually amplify our main findings. See also p.22.

- 2) We also provide a sensitivity analysis on misreporting: How much underreporting of true transfer receipt by high-consumption people would be required to zero out our self-targeting coefficients. The answer is: an implausible rate of underreporting (more than half of true recipients don’t report it), as we discuss on p.22.

We hope these analyses “leave no stone unturned” on transfer misreporting as a threat to our analysis. If the Reviewer has specific language or tests in mind they would like to see, we welcome suggestions.

- 18. “Isn’t a slightly odd feature of the adjustment to automatic enrollment that your proposed policy alternative punishes people at income levels with more self-selection more? That suggests that it may not be the welfare-maximizing adjustment. Maybe it is still the most reasonable adjustment, but it is a policy that seems unlikely to happen in practice, so it may be useful to have a short discussion of how these choices affect welfare analyses and how one could develop similar criteria for specific policy proposals.”**

We thank the Reviewer for raising this question. We understand it as follows: In order to give \$1 on average to people with an income y , a voluntary transfer with a take-up rate $M(y)$ must take $M(y)/(1-M(y))$ dollars from eligible nonrecipients at that income. This ratio implies that eligible nonrecipients are “punished” more for non-take-up when similar-income people have high take-up than such nonrecipients would be at points in the income distribution with low take-up. Given that take-up rates decline in income, this implies that relatively low-income eligible recipients are “punished” harder than relatively high-income eligible nonrecipients.

However, every distributionally-neutral reform has this property. We see income-neutrality as an important property of our reform, so we do not conflate “incidental” redistribution with the reform (Kaplow, 2011). Overall, we are happy to add a footnote summarizing the above discussion if helpful!

- 19. “P.22: I think throughout this discussion, “reducing” and “increasing” progressivity are flipped. Doesn’t reducing progressivity mean that transfers increase less with income? So going automatic increases transfer progressivity, since it moves receipt higher up the income distribution? Just harmless wording.”**

Section 4 has been cut in its entirety, following comments from Reviewer 1. As a result, the text relevant to this comment is removed from the paper.

20. **“Given the empirical problems, I find the conclusions of the empirical application to be stated rather strongly in some cases. They clearly point in one direction, but given the data and implementation issues above, I am less confident about conclusions regarding specific programs, how they compare and what the ideal policy would be than you seem to be. I am not sure whether and to what extent it is necessary to rephrase things more cautiously, but I see the application more as a proof of concept and usability (including proof of how much we need to improve the data) than as something I would use for policy here and now.”**

We thank the Reviewer for raising this point, which we address in their Comment 16.